A Field Report on

Physicochemical and Geomorphological Profiling of the Sylhet: An In-Situ Investigation of Pedological, Hydrological, and Microclimatic Variables in Lawachara, Jaflong, Volaganj, and Madhabkunda in Sylhet

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Abstract

This report aims to describe and analyze the physical environment of Sylhet division. The fieldwork conducted by second-year students from the Department of Environmental Science and Geography at Islamic University, Bangladesh. The study set out to observe four different locations in the northeast of Bangladesh. The goal was to study the area's physiography, hydrology, soil properties, atmospheric (weather) parameters, geomorphology using both direct measurements and informed interpretations.The study area is located in the north-east region of Bangladesh. To be specific on the southern edge of the Shillong Plateau within the Surma Basin. There the land isn't flat. Alternately it rises and falls, with lots of floodplains, hills, and valleys. Elevations vary from 50 feet up to 300 feet above mean sea level (MSL). Rivers were mainly the Piain River and Gowain River that develop a dendritic drainage system throughout the landscape. Due to the area's humid subtropical climate, it delivers moderate temperature, heavy monsoon rains and elevated humidity. The area is close to the Dauki Fault Region. This case study has significantly dictated us to know the regional geomorphology and sediment deposition. During this fieldwork, we collected air temperature, relative humidity, atmospheric pressure, wind direction and wind speed. We were using a portable weather station. Also we took soil and water samples for both physical and chemical analysis. In Volaganj and Madhabkundo, we gathered unique rocks and minerals for identification and further lab testing, indicating the region's geological diversity and how the river stream flow and hill processes shaped mineral and sediment deposits. We checked the quality of water by measuring water pH, electrical conductivity, salinity, and total dissolved solids (TDS). Portable Tools like the Soil Moisture and pH Meter, the Soil Electrical Conductivity Meter are used to record soil moisture, temperature, and electrical conductivity.

Finally, this field investigation provided hands-on real world experience. It helped to link the theory to the real-world environment of Sylhet and learn how to collect and represent of environmental and geographical data.

Table of Contents

CHAPTER 01: INTRODUCTION 1

1.1 Background 1
1.2 Aims and Objectives 2

CHAPTER 02: STUDY AREA 3

2.1 Environmental and Geographical Characteristics 3

CHAPTER 03: MATERIALS AND METHODS 5

3.1 Materials 5
3.2 Experiment Methods 8

CHAPTER 04: RESULTS AND DISCUSSIONS 10

4.1 Observation by Route 10
4.2 Area Based Discussion 11
4.2.1 Lawachara National Park *12*
Micro-meteorological Condition 12
Air Quality 16
Water Quality 17
Soil Condition 20
Overall Environmental Condition: 23
4.2.2. Jaflong Zero Point *25*
Micro-meteorological Condition 25
Air Quality 27
Water Quality 29
Soil Condition 32
Overall Environmental Condition: 34
4.2.3 Sada Pathor, Bholaganj *35*
Micro-meteorological Condition 35
Air Quality 37
Water Quality 39
Soil Condition 42
Overall Environmental Condition: 44
4.3 Site Based Observation 45
4.3.1 Lawachara National Park *45*
Flora 45
Fauna 48
4.3.2 Jaflong Zero Point *49*
Geomorphology and Geography 49
Rocks and Minerals (Jaflong) 51
4.3.3 Sada Pathor, Volaganj *53*
Geomorphology and Geography 53
Rocks and Minerals (Sada Pathor) 55
4.3.4 Madhabkunda Eco Park *57*
Geomorphology and Geography 57
Rocks and Minerals (Madhabkunda Eco Park) 59

CHAPTER 05: CONCLUSION AND RECOMMENDATIONS 61

5.1 Conclusion 61
5.2 Limitations of the Study 62
5.3 Recommendations 63

CHAPTER 06: REFERENCES 64

CHAPTER 07: APPENDICES 65

7.1 DATA TABLE 65

7.2 PICTURES FROM FIELD WORK 66

List of Tables

TABLE 1: AREA BASED DISCUSION ____ 11
TABLE 2: TEMPERATURE IN LAWACHARA ____ 12
TABLE 3 RH, LAWACHARA ____ 13
TABLE 4: PRESSURE IN LAWACHARA ____ 14
TABLE 5: WS & WD IN LAWACHARA ____ 15
TABLE 6: AIR QUALITY IN LAWACHARA ____ 17
TABLE 7 WATER QUALITY, LAWACHARA ____ 19
TABLE 8 WATER EC, LAWACHARA ____ 20
TABLE 9 SOIL EC,PH,MOISTURE, LAWACHARA ____ 23
TABLE 10 ATMOSPHERE CONDITION, JAFLONG ____ 25
TABLE 11 AIR QUALITY ____ 27
TABLE 12 WATER QUALITY, JAFLONG ____ 31
TABLE 13 SOIL QUALITY, JAFLONG ____ 32
TABLE 14 ATMOSPHERE CONDITION, SADA PATHOR ____ 37
TABLE 15 WATER QUALITY, SADA PATHOR ____ 40
TABLE 16 SOIL QUALITY, SADA PATHOR ____ 44
TABLE 17 DATA TABLE ____ 65

List of Figures

Figure 1: Study Area map ____ 3
Figure 2: Materials Used ____ 5
Figure 3: Temperature Graph in Lawachara ____ 12
Figure 4: Temperature in Lawachara ____ 12
Figure 5: RH in Lawachara ____ 13
Figure 6: RH in Lawachara ____ 13
Figure 7: Pressure in Lawachara ____ 14
Figure 8: Pressure in Lawachara ____ 14
Figure 9: PM 10 in Lawachara ____ 16
Figure 10: PM 2.5 in Lawachara ____ 16
Figure 11: PM Graph ____ 16
Figure 12 Water Salinity, Lawachara ____ 18
Figure 13 TDS, Lawachara ____ 18
Figure 14 Water Quality Graph, Lawachara ____ 18
Figure 15 Water pH, Lawachara ____ 18
Figure 16 Water EC Graph, Lawachara ____ 19
Figure 17 Water EC, Lawachara ____ 19
Figure 18 Soil EC,pH,Moisture Graph, Lawachara ____ 22
Figure 19 Soil EC, Lawachara ____ 22
Figure 20 Soil pH, Lawachara ____ 22
Figure 21 Soil Moisture, Lawachara ____ 22
Figure 22 Pressure, RH, Temp., Jaflong ____ 26
Figure 23 Temperature, Jaflong ____ 26
Figure 24 Presure, Jaflong ____ 26
Figure 25 RH, Jaflong ____ 26
Figure 26 PM 2.5, Jaflong ____ 28
Figure 27 PM 10, Jaflong ____ 28
Figure 28 PM Graph, Jaflong ____ 28
Figure 29 Water EC, Jaflong ____ 30
Figure 30 TDS, Jaflong ____ 30
Figure 31 Water pH, Jaflong ____ 30
Figure 32 water Salinity, Jaflong ____ 30
Figure 33 Water Quality Graph, Jaflong ____ 31
Figure 34 Soil pH, Jaflong ____ 33
Figure 35 Soil Moisture, Jaflong ____ 33
Figure 36 Soil EC, Jaflong ____ 33
Figure 37 Soil Quality Graph, Jaflong ____ 33
Figure 38 Pressure, Sada Pathor ____ 36
Figure 39 Temp. RH, Pressure, Sada Pathor ____ 36
Figure 40 RH, Sada Pathor ____ 36
Figure 41 Temp., Sada Pathor ____ 36
Figure 42 PM Graph, Sada Pathor ____ 38
Figure 43PM 2.5, Sada Pathor ____ 38
Figure 44 PM 10, Sada Pathor ____ 38
Figure 45 Water Quality, Sada Pathor ____ 40
Figure 46 Water EC, Sada Pathor ____ 41
Figure 47 Water pH, Sada Pathor ____ 41
Figure 48 Water Salinity, Sada Pathor ____ 41
Figure 49 TDS, Sada Pathor ____ 41
Figure 50 Soil Moisture, Sada Pathor ____ 43
Figure 51 Soil Quality Graph, Sada Pathor ____ 43

Figure 52 Soil pH, Sada Pathor 43
Figure 53 Soil EC, Sada Pathor 43
Figure 54 Musa acuminata 45
Figure 55 Melastoma malabathricum 45
Figure 56 Trametes versicolor 46
Figure 57 Gmelina arborea 46
Figure 58 Tectaria sp. 46
Figure 59 Tectona grandis 47
Figure 60 Dipterocarpus turbinatus 47
Figure 61 Phrynium pubinerve 47
Figure 62 Amomum sp. 48
Figure 63 Rock Sample 1 51
Figure 64 Rock Sample 2 51
Figure 65 Rock Sample 3 51
Figure 66 Rock Sample 4 52
Figure 67 Rock Sample 5 52
Figure 68 Rock Sample 6 55
Figure 69 Rock Sample 7 55
Figure 70 Rock Sample 8 55
Figure 71 Rock Sample 9 56
Figure 72 Rock Sample 10 56
Figure 73 Rock Sample 11 59
Figure 74 Rock Sample 12 59
Figure 75 Rock Sample 13 59
Figure 76 Rock Sample 14 60
Figure 77 Rock Sample 15 60

Chapter 01: Introduction

1.1 Background

As part of the academic curriculum at the Department of Environmental Science and Geography, Islamic University, Bangladesh, the second year students took part in a fieldwork program under the course called "Observation Survey: Physical Environment Lab" (Course Code: ESG-2209). The main goal for this fieldwork was providing direct experience and practical knowledge of the physical environment of Sylhet division. This study proposed to sampling survey places throughout northeast Bangladesh that focuses on things like microclimatic differences, geological structure, geomorphic processes, and the quality of soil, water, and air.

In geological terms, the study area is located in the Sylhet Trough, a rapidly sinking sub-basin in the larger Bengal Basin. This area sits on the southern edge belonging to the Shillong Plateau and derives much of its geological character from the tectonic activity connected to the Himalayan orogeny. The landscape here mostly takes shape from the Dauki Fault system, a major boundary separating the uplifted Precambrian rocks of the Shillong Plateau from the thick Tertiary sedimentary layers of the Surma Basin (Uddin & Lundberg, 2004; Islam et al., 2011). Because of its complicated tectonic history, this region shows up as rolling terrain with different elevations a landscape full of marked anticlines, floodplains, and valleys. Plus, the active river network, led by rivers like the Piain and the Gowain, forms a dendritic drainage pattern that constantly brings heavy loads of sediment from the Meghalaya hills into Bangladesh, shaping how land and deposits build up in the region (Johnson & Alam, 1991).

The Sylhet region has plentiful natural resources and ecological diversity, because of its humid subtropical climate having moderately mild temperatures and strong monsoon rains. But honestly, this benefit is shrinking fast due to unchecked land-use changes, natural resource exploitation, and climate changes. Important environmental sites are up against serious trouble: soil is breaking down, rivers are changing course, and air and water quality are dropping off. In places like Jaflong and Bholaganj, heavy stone and sand extraction from riverbeds is throwing off the river's natural flow, making the water murkier and speeding up riverbank erosion (Hossain et al., 2019). On top of that, human pressures from chopping down forests in areas like Lawachara to growing tourism and expanding industries are slicing up local habitats and throwing the fragile ecological balance way off in these sensitive zones (Rahman et al., 2020).

Even with the region's huge geological significance and the clear ecosystem decline, there's still a real need for ongoing, hands-on monitoring that pulls together soil, water, and climate data. That's why this field investigation tries to connect what's taught in the classroom with what students see out in the world, by offering a practical look at geographical data to help make sense of the rapid changes happening in Sylhet's physical landscape.

1.2 Aims and Objectives

The main objective of this fieldwork was to observe the physical environment of Sylhet, as well as the interconnections among air, water and soil, micro climate, and geomorphological features. This study helps to track variations in fundamental environmental parameters through the study area. Our goal was to solve the puzzle how these elements interact to shape the overall ecosystem.

The specific objectives of the study were as follows:

1. Geomorphologic Study: This study teaches about landforms and landscapes of northeastern Surma Basin. It also describes the boundary of the Shillong Plateau, foot hill and active river channels.
2. Pedological (Soil) Assessment: This study observes and analyzes soil characteristics, including texture, pH, and moisture and electrical conductivity.
3. Water Quality Assessment: This study evaluates the water quality of selected sites by measuring Total Dissolved Solids (TDS), salinity, electrical conductivity (EC), and pH.
4. Microclimatic and Air Quality Assessment: This study records and compares regional atmospheric conditions by measuring temperature, relative humidity, atmospheric pressure, and wind speed as well as air particulate matter PM 2.5 and PM 10.
5. Technical Proficiency: This study helps to develop practical knowledge of specialized environmental monitoring equipment, including portable weather stations, digital particle counter meter, Water Series Tester, Soil EC Meter and real-time soil pH and moisture meters.
6. Ecosystem Interconnectivity Assessment: To investigate the complex, real-world interdependencies between regional landforms, climate variables, pedology, and vegetation dynamics across distinct landscape types, such as upland forests (Lawachara), waterfalls (Madhabkundo), and heavily modified river basins (Jaflong and Bholaganj).
7. Academic and Methodological Development: To bridge theoretical classroom concepts with practical application, thereby refining the essential skills required for rigorous spatial observation, empirical data collection, and the scientific interpretation of geographical datasets.

Chapter 02: Study Area

Our study was held at four specific locations Jaflong, Bholaganj, Lawachara, and Madhabkundo. This reflects the diverse physical, geological, environmental and ecological image of the Sylhet division. Within the Surma Basin, these spots were highly informative for realizing the region's up and down topography, hydrology, and biodiversity. In addition, the transit routes between these locations played an important role during the fieldwork. It blesses us to observe the shifting gradient of vegetation, land use, and the broader humid subtropical environment.

1. Lawachara National Park
2. Jaflong Zero Point
3. Volaganj Sada Pathor and
4. Madhabkundo Eco Park

2.1 Environmental and Geographical Characteristics

Sylhet Division is located in the northeastern corner in Bangladesh, from 24°30′N to 25°15′N latitude and 91°38′E 92°30′E longitude. To the north, east, and south, it shares boundaries with India's Meghalaya, Assam, and Tripura a landscape set apart from much of Bangladesh due to its unique character within the active Surma Basin. The division's position at the southern edge within the Shillong Plateau shapes its geology in distinctive ways from the country's more typical flat terrain.

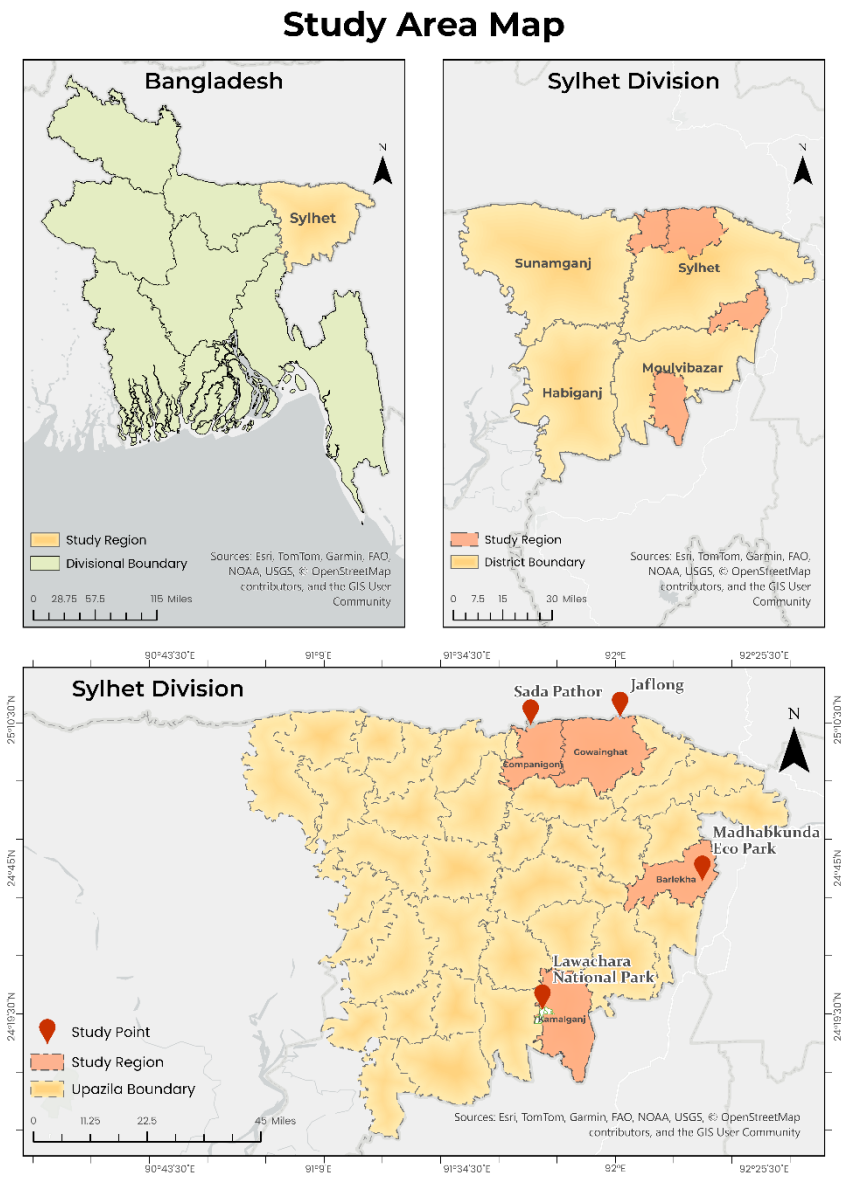


Figure 1: Study Area map

Sylhet's landforms shifting from one place to another. Undulating elevations break up flat floodplains and sprawling wetlands known as haors. In the center and west, riverine plains and haor basins are predominant. The east and north, though, are cut by hill ranges that reach out from the Jaintia and Tripura Hills. During fieldwork, that variety became obvious standing near the Shillong Plateau, the land rises from 50 up to 300 feet above mean sea level. All together, the hills wrap Sylhet into a kind of basin, walled in on three sides, with the Dauki Fault marking its southern edge.

Water shapes everything here. The division's drainage network sprawls wide and deep, with the Surma-Kushiyara river system at its heart. These rivers start as the Barak River in India, and with tributaries like the Piain and Gowain noted in surveys they spread out in a branching, dendritic pattern. This river system doesn't just shape the land it carries heavy sediment from the hills and lays it out downstream, building up tidal flats and twisting, braided channels. When the monsoon rains hit, those same streams spill over, flooding floodplains and haors until the whole landscape changes.

Sylhet Division's climate is thoroughly humid and subtropical; few places in Bangladesh see more rain. The southwest monsoon arrives with long spells of rainfall and sticky air.

- Mean yearly precipitation: 3,500 4,500 mm
- Temperature range:
  - Summer: 30 - 35°C (sometimes above 40°C in overused or altered landscapes)
  - Winter: 10 - 15°C

Soils here mostly come from modern alluvial deposits and older sedimentary layers. Down in the haors and floodplains, clay and silt make the soil rich but they flood easily, and water can linger for weeks. Up in the hills, brown soils that field tests pegged at slightly acidic (pH 5.3 to 6.8) and not as fertile. Underneath, the Sylhet Trough holds more than just earth there's limestone (like Sylhet Limestone), sand, sedimentary rock, and valuable natural gas reserves.

The natural flora of Sylhet differs place to place. Over the years, logging, mining, farming, and tourism have carved up. But still shrunk the natural forest is very important for the ecology.

- Elevated terrains still hold sections of lush tropical evergreen and semi-evergreen forest, that trap moisture and support unique microclimates.
- Reserves like Lawachara National Park shelter much of the region's remaining biodiversity, though they face mounting threats from encroachment.
- Down in the haors and floodplains, reeds, tall grasses, swamp forests Ratargul, for example and aquatic plants built for land that floods and dries with the seasons.

Chapter 03: Materials and Methods

3.1 Materials

Various type of portable devices, instruments were used during the field work to collect, store and analyze data. Here is the list of used instruments.

1. Pen
2. Pencil
3. Notebook
4. Tissue
5. Water Sample Collection Bottles
6. One time glass and plastic bags
7. GPS Receiver
8. Water Series Tester
9. Soil pH and Moisture Meter
10. Soil Conductivity Meter
11. Portable Weather Station
12. Portable Particulate Counter
13. Mohs Hardness Tools kit



Figure 2: Materials Used

3.2 Experiment Methods

Working Procedure of the GPS Receiver

Collecting spatial data with the GPS receiver was a step-by-step process. First, I turned on the device and set it up in an open spot with a clear view of the sky so it could grab satellite signals without any trouble. I keep it still for a few minutes. And wait to lock onto at least 4 satellites. Then let the location data stable, so I'd get more accurate results. After over 120 seconds, I took the coordinates. After wrapping up data collection at a site, I made sure to shut the unit down properly.

Working Procedure of the Portable Soil pH and Moisture Meter

At first I had taken the soil pH and moisture reading, using a portable soil pH and moisture meter. Before starting, I properly cleared the soil surface and ensured no loose bits, rocks, or heavy organic stuff on the electrodes. so the readings would be corrected. I dipped the metal probe into the topsoil, going down about 5 cm to 10 cm, and made sure it stayed tight against the top soil. Then I waited for the display to steady, then logged the numbers. Then I let the probe sit for a moment to let the chemical reaction on the sensor even out. Then took down the pH and moisture (%) values. After finishing all these, I pulled off the probe out carefully and cleaned it dry to keep different samples from mixing.

Working Procedure of the Portable Water Series Tester

I used a portable series tester to measure pH, Electrical Conductivity (EC), Salinity, and Total Dissolved Solids (TDS) of sample water. First thing, I took off the protective cap and switched the device on. Then I dipped the sensor probe straight into the collected water sample until it hit the immersion line marked by the manufacturer. I gently stirred the tester in the water, making sure there weren't any air bubbles stuck around the electrode, and waited for the digital display to settle down. Then I switched through each measurements like pH, EC, Salinity, and TDS by pressing button. Then I wrote down the stable readings. When I finished at each test, I cleaned the electrode with clean water, wiped it with tissue and put the protective cap back on.

Working Procedure of the Portable Particulate Matter Counter

I used a portable digital particulate matter counter to measure air quality. First, I turned it on and checked the battery to make sure it had enough charge. I set the device at about 1.2 to 1.5 meters above the ground right at breathing height so the readings would reflect what people actually breathe in. I let the sensors warm up for a bit. After that, I remained the device steady in an open area and made sure nothing blocked the air intake not even my hands. Then I set time. At the moment the screen showed stable readings, I wrote down the PM concentrations for PM2.5 and PM10 in notebook. After I collected data for the location, I stopped measuring.

Working Procedure of the Portable Soil Electrical Conductivity (EC) Meter

To measure soil EC, I used a portable digital soil EC meter. I started by cleaning the soil surface. I turned the device on and pushed the sensor probe into the topsoil, about 5 to 10 centimeters deep. I made sure the electrodes had good contact with the soil. I held the device steady while the sensors adjusted. Then, I recorded the EC value in µS/cm and Temperature. After each reading, I took out the probe carefully and cleaned it with tissue.

Working Procedure of the Portable Weather Station

First, we picked an open spot no trees, with wind readings or sunlight. We anchored the tripod really well and mounted the weather station on top. After that, we used the station's bubble level and adjusted the tripod legs until it sat perfectly flat. Next, we checked every sensor temperature, humidity, wind speed, wind direction, and atmospheric pressure to make sure they were connected right and nothing blocked them. Then we powered everything on and waited a short while. I gave sensors time to adjust to the outside air. After that I started recording the weather data at set intervals, always noting the exact date and location When we finished recording.

Chapter 04: Results and Discussions

4.1 Observation by Route

The observation journey started at Islamic University, right there in the western alluvial plains of Bangladesh. This area's mostly farmland think wide stretches of rice, jute, and whatever seasonal veggies are in rotation. We see a patchwork of active crop fields, some homestead trees, scattered ponds, and a network of irrigation canals. People don't live packed in here. Settlements are rural, spread out across the floodplain, and that's mostly because of the reliable Gorai River system.

Marching northeast toward the Sylhet region, the landscape really starts to change. The rolling hills have given way to plain floodplains, and we arrive at Reserved Forest Lawachara National Park. The forest situated rectangularly in the Surma River basin. It has little forest streams threading through the shade. The land use is all about biodiversity conservation and ecology friendly tourism. A thick canopy of evergreen trees covers the ground. There is leaf litter piles up. Also the air feels very cooler and very humid. Hardly anyone lives in this dense forest and settlements are hardly visible.

The path then leads northeast to Jaflong, set on the Bangladesh India border at the foot of the Shillong Plateau. Piain River running down from the Meghalaya Hills, loaded with sediment and minerals. The river's clear and shallow, flowing over broad, rocky beds littered with boulders. Here, stone and boulder extraction shapes the land, with a bit of tourism on the side. The area shows off heavily altered river channels, patches of bare mineral deposits, and not much vegetation by the water. Farming is pretty much out of the question the ground's too rocky and uneven for that.

Continuing along the border region, the route reached Bholaganj. Similar to Jaflong, this area is heavily shaped by active river systems flowing from the upland plateau, but it exhibits a much more severe degree of environmental and anthropogenic stress. The Land use is gighly resource-oriented and severely dominated by heavy machinery-based stone and sand extraction processes. The only visible land cover consists of vast exposed, dry sandy riverbeds, large boulder tracts, and highly degraded, sparsely vegetated floodplains. The sorrounding environment was notably harsher. There was high amount of suspended dust. Also high surface heat reflecting off the barren, extracted sandy terrain.

The trip keeps moving to Madhabkundo. Now in a genuinely hilly place, right on the edge of the regional plateau. The steep land here is cut by fast-moving streams, especially the one that feeds Madhabkundo Waterfall. Now the land is covered with thick hilly forests and that is exposed rocky surfaces. Also there was narrow, deep stream channels. Unlikely the flat agricultural plains back in west. this rough, energetic river landscape feels completely different to us.

After completing the environmental assessments at these four primary locations in the Sylhet region, the return journey moved southwest.

Finally, the expedition concluded with the return to Islamic University in Kushtia, where the landscape seamlessly transitioned back to the stable, flat agricultural floodplains of the western region.

4.2 Area Based Discussion

We chose an area-based approach because the study region covers a wide mix of ecological zones thick forests in some spots, open riverbanks in others. If we merged all this information, we'd end up blurring the important details for each place. To stay accurate, we recorded primary environmental variables like meteorology, air, water, and soil by gathering three individual samples from each sampling point. We kept the data for each site separate and visualized it to spot local patterns. Then we compared our findings to established international standards, such as WHO guidelines. This gave us a clear, accurate picture of the unique environmental health across all the different ecosystems.

Table 1: Area Based Discusion

Feature	Area-Based Discussion	Parameter-Based Discussion
Primary Focus	The holistic ecosystem of a specific location.	A specific environmental variable across multiple locations.
Structure	Analyzes all parameters (soil, water, air, weather) site by site (e.g., Lawachara, then Jaflong).	Analyzes one parameter (e.g., pH, Temperature) across all sites simultaneously.
Key Advantage	Highlights how different environmental factors interact to create a unique local microclimate.	Easily highlights regional trends, spatial gradients, and extreme outliers for a single metric.
Contextual Depth	High. Preserves the complex ecological reality of an isolated environment.	Low for local ecosystems, but high for regional comparative analysis.
Best Used For	Evaluating overall site health, ecological profiling, and localized conservation planning.	Regional mapping, compliance monitoring, and identifying geographical pollution spread.

4.2.1 Lawachara National Park

Micro-meteorological Condition

Temperature:

The temperature at reserved forest of Lawachara National Park is characterized by a moderate and highly stable microclimate. The surface temperatures remained relatively cool and steady. Temperature remains between 28.4°C and 30.3°C most of the time. This is for its mixed evergreen forest. That acts as a natural thermal insulator. By blocking direct solar radiation and promoting high rates of cooling evapotranspiration process, the forest prevents extreme heat buildup. That results in a significantly cooler and more insulated environment compared to the bare, heavily modified landscapes found elsewhere in the Sylhet region.

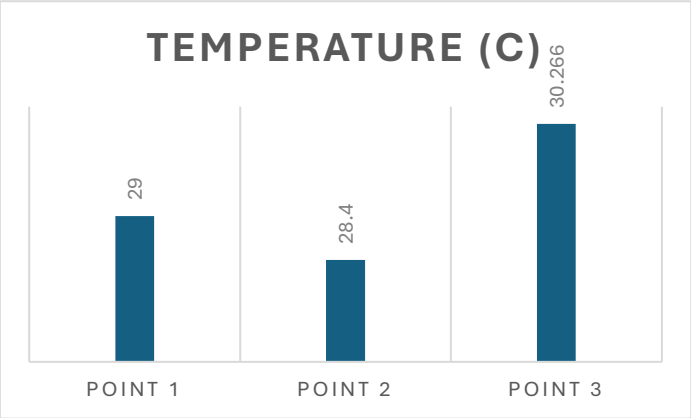


Figure 3: Temperature Graph in Lawachara

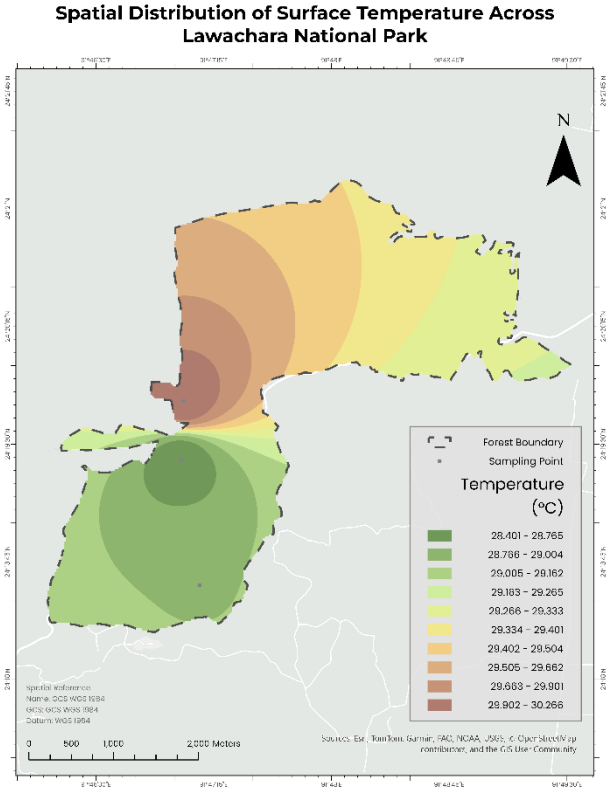


Figure 4: Temperature in Lawachara

Table 2: Temperature in Lawachara

Location	S\N	Latitude	Longitude	Temp. (C)	Avarage
Lawachara National Park	Point 1	24.31	91.786	29	29.2
	Point 2	24.32332	91.784013	28.4	
	Point 3	24.329598	91.784321	30.266	

Humidity:

Lawachara National Park stays incredibly humid relative humidity sits between 74.8% and 81.9% at every spot researchers check. That's not just a quirk; the park's thick, layered forest canopy holds onto moisture, making it hard for sunlight to dry things out. The trees and plants keep releasing water vapor into the air, so humidity rarely drops. This moisture-loaded, sheltered atmosphere is central to the park's tropical ecosystem, providing the conditions its rich variety of species need to thrive.

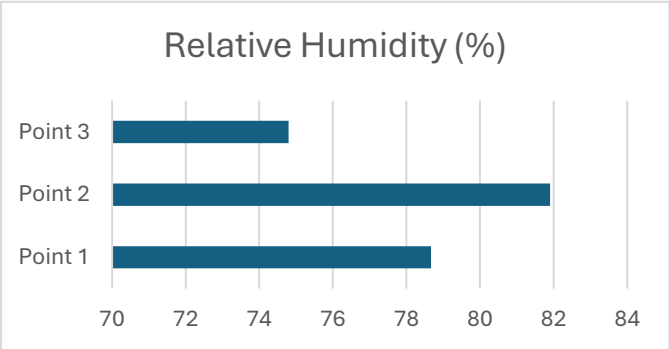


Figure 6: RH in Lawachara

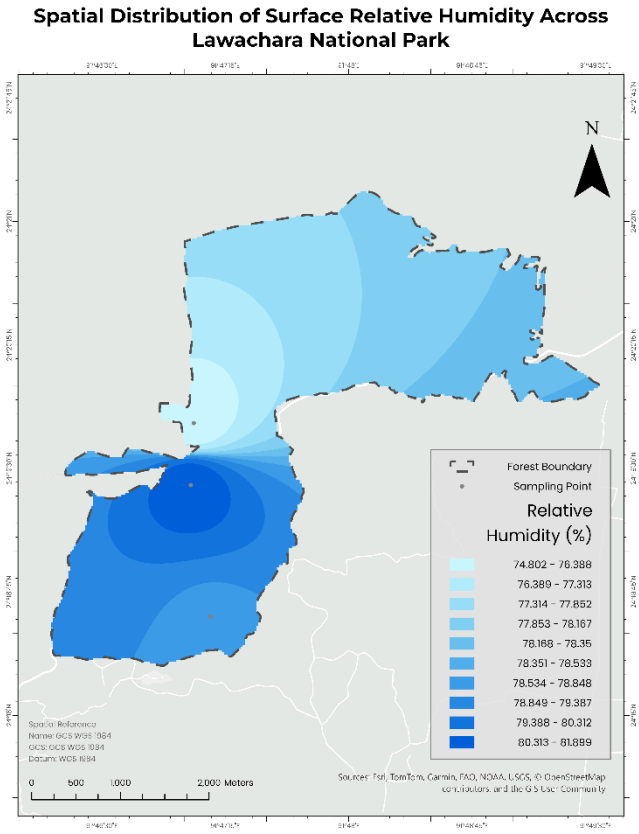


Figure 5: RH in Lawachara

Table 3 RH, Lawachara

Location	S\N	Latitude	Longitude	RH (%)	Avarage
Lawachara National Park	Point 1	24.31	91.786	78.67	78.5
	Point 2	24.32332	91.784013	81.9	
	Point 3	24.329598	91.784321	74.8	

Atmospheric Pressure:

Atmospheric pressure in Lawachara National Park barely shifted between sampling points. The pressure was between 1003.0 and 1005.8 hPa. During the study the weather was calm. The thick forest works like a barrier, blocking out strong winds. That's clear from the zero wind speed recorded. This natural shield keeps pressure changes minimal, helping the park maintain an impressively steady microclimate.

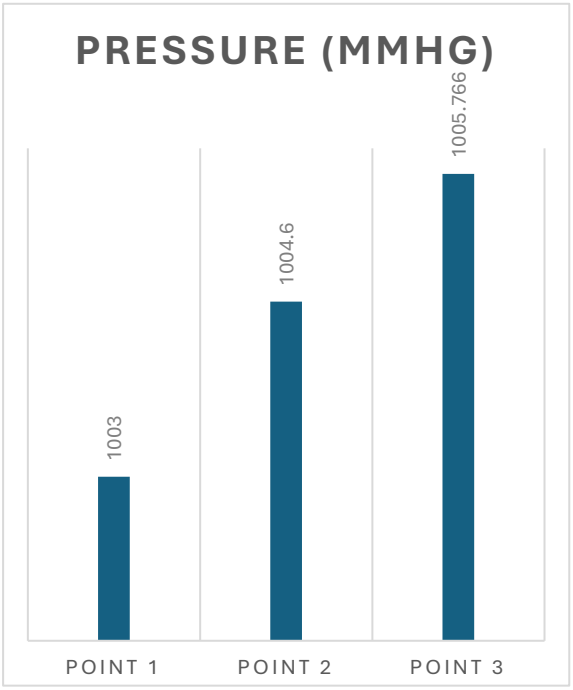


Figure 8: Pressure in Lawachara

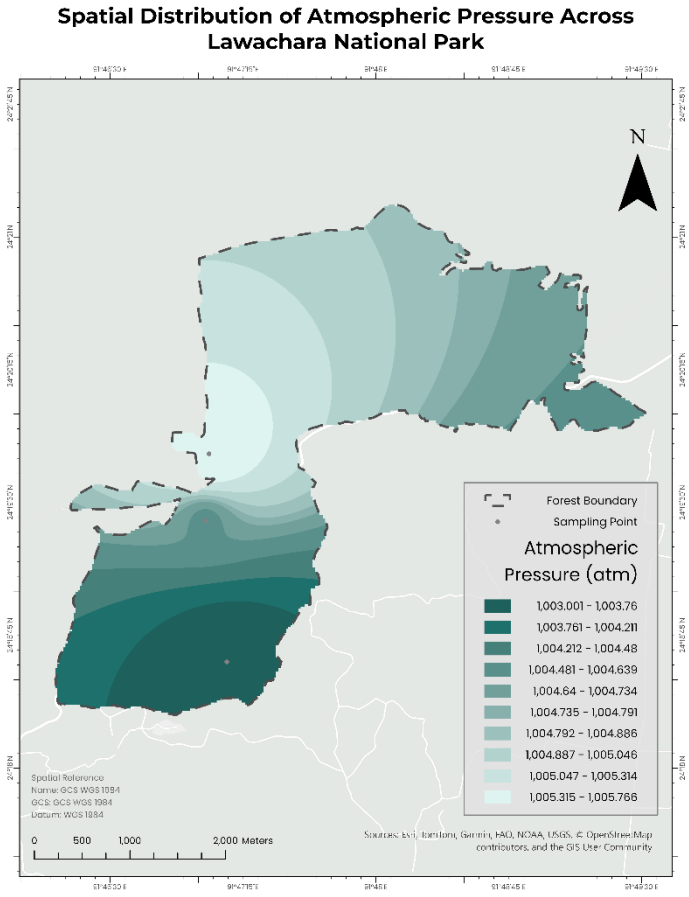


Figure 7: Pressure in Lawachara

Table 4: Pressure in Lawachara

Location	S\N	Latitude	Longitude	Pressure (mmHg)	Avarage
Lawachara National Park	Point 1	24.31	91.786	1003	1004.5
	Point 2	24.32332	91.784013	1004.6	
	Point 3	24.329598	91.784321	1005.766	

Wind Speed and Direction:

At Lawachara National Park, the wind didn't budge not even a whisper at any of the three sampling spots. Wind direction was impossible to measure. Dense blanket of trees form a wall. That blocks wind before it comes close to touching the ground. This brakes on any air movement, which keeps the atmospheric pressure steady and locks in moisture near the ground. A calm, insulated patch of forest that feels almost untouched by what's stirring outside.

Table 5: WS & WD in Lawachara

Location	S\N	Latitude	Longitude	WS (m/s)	WD
Lawachara National Park	Point 1	24.31	91.786	0	NA
	Point 2	24.32332	91.784013	0	NA
	Point 3	24.329598	91.784321	0	NA

Overall Weather Condition:

Lawachara National Park feels calm and wrapped in humidity almost like its own little world. Temperatures hover around a warm 29°C, but what really stands out is the humidity, which climbs up to nearly 82%. It's thick, lush air. There's no wind on the ground at all, and the atmospheric pressure stays steady, making the whole place feel quiet, almost insulated from the outside. All of this traces back to the dense evergreen canopy overhead. It holds in moisture, keeps out strong breezes, and regulates temperature, shaping a microclimate that's both unique and thoroughly sheltered a true sanctuary for the park's plants and animals.

Particulate Matter (PM 2.5 and PM 10):

Particulate matter levels at Lawachara National Park stay impressively low. PM 2.5 values barely shift, sitting between 5.30 and 7.30 µg/m³ at the different sampling spots. The larger particles, measured as PM 5 in this area, don't go above 7.67 µg/m³ most readings hover just above 5.70 µg/m³. With numbers like these, it's clear there's hardly any dust, soot, or human-made aerosols floating around. The forest's dense, layered canopy works almost like a living filter. All those overlapping leaves catch and hold particles out of the air, keeping pollution from seeping in from beyond the park.

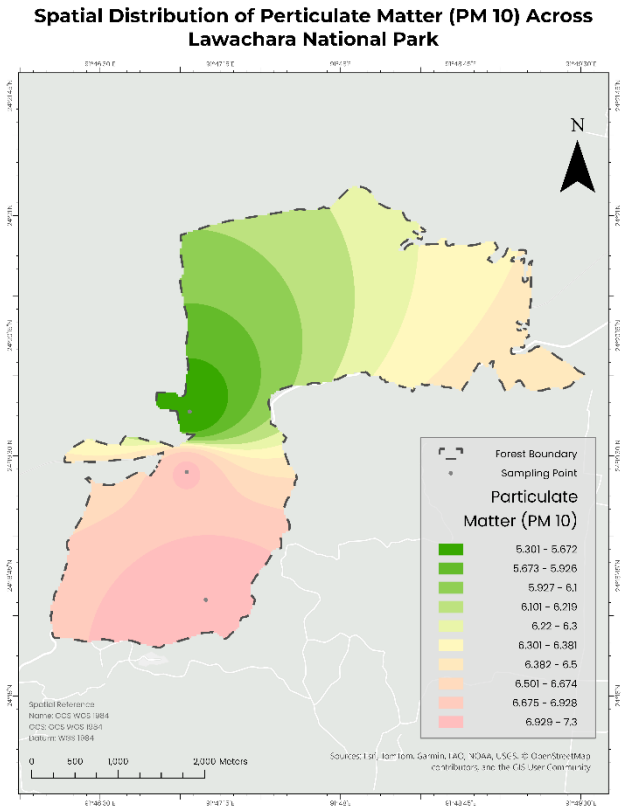


Figure 9: PM 10 in Lawachara

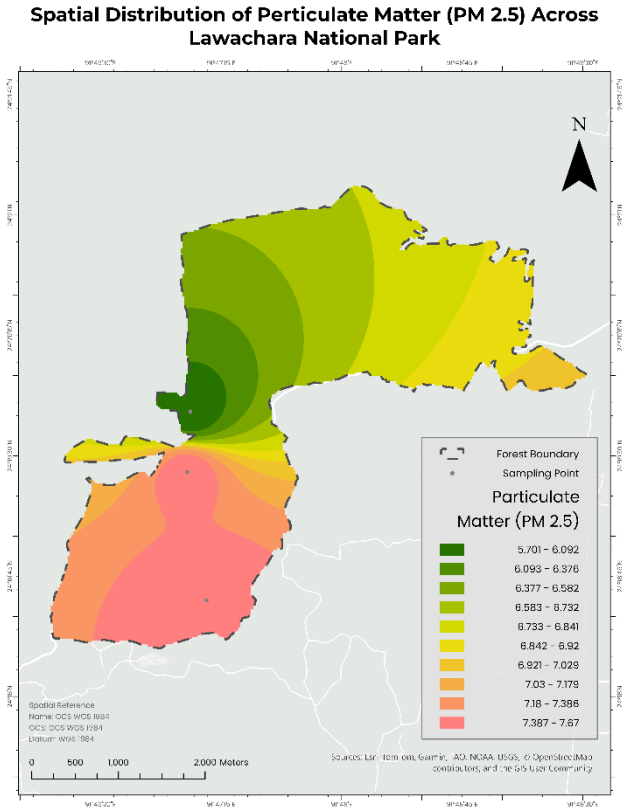


Figure 10: PM 2.5 in Lawachara

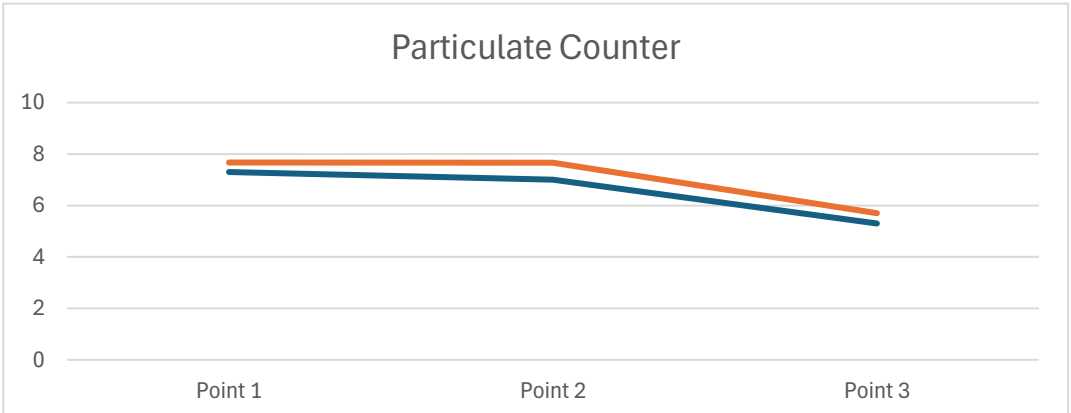


Figure 11: PM Graph

Table 6: Air Quality in Lawachara

Location	S\N	Latitude	Longitude	PM 2.5	PM 5
Lawachara National Park	Point 1	24.31	91.786	7.3	7.67
	Point 2	24.32332	91.784013	7	7.667
	Point 3	24.329598	91.784321	5.3	5.7

Overall Air Quality:

Lawachara National Park has remarkably clean air. With hardly any particulate matter and no industry or mining nearby, pollution just isn't a problem here. The thick evergreen canopy does a lot of the work blocking dust and acting as a natural air filter. Thanks to this, the park gives its plants and animals a rare chance to thrive in pure, undisturbed air. For the region's wildlife, it's a real sanctuary.

Water Quality

Total Dissolved Solids (TDS):

Lawachara National Park's streams are impressively pure. In the sample water of here the Total Dissolved Solids (TDS) remains between 19.34 and 20.70 mg/L. This is far below the standards. For reference, the World Health Organization rates anything under 300 mg/L as excellent for drinking. If it is less than 100 mg/L, that's practically untouched, barely any minerals at all. Lawachara's readings go even lower. This isn't just a number on a lab report it means the water slips through the forest with hardly any dissolved salts, sediment, or heavy minerals. The roots and leaf litter do their job, trapping runoff and stopping minerals from leaching into the streams. It is a glimpse of a hydrological system that's still intact, still working as it should.

Water Salinity:

Water salinity at Lawachara National Park is surprisingly low between 24.40 mg/L and 26.60 mg/L at all three sampling sites. That's way below the 500 mg/L mark set by the World Health Organization for excellent quality freshwater. So, these streams really are pure freshwater. No traces of farm chemicals, industrial waste, or salt from the rocks. It shows the park has a nearly untouched, rain-fed water system, with the forest's soil quietly filtering everything before it reaches the streams.

Water pH:

Water in Lawachara National Park is a bit on the acidic side. Measurements at three spots show pH levels between 5.58 and 6.03. That's lower than what the World Health Organization recommends for drinking water 6.5 to 8.5 is their sweet spot. But, that's not surprising out here. Forests like this have tons of leaf litter breaking down, which releases natural acids like humic and fulvic acids into the soil and streams. That's why the water's pH drops. So, even though it doesn't quite fit the guidelines for

drinking water, these slightly acidic conditions are normal and perfectly healthy for a forest like Lawachara.

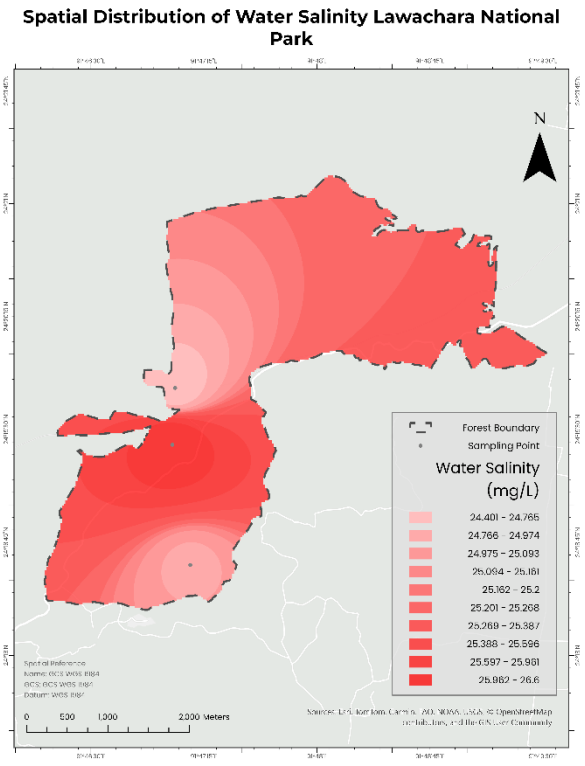


Figure 12 Water Salinity, Lawachara

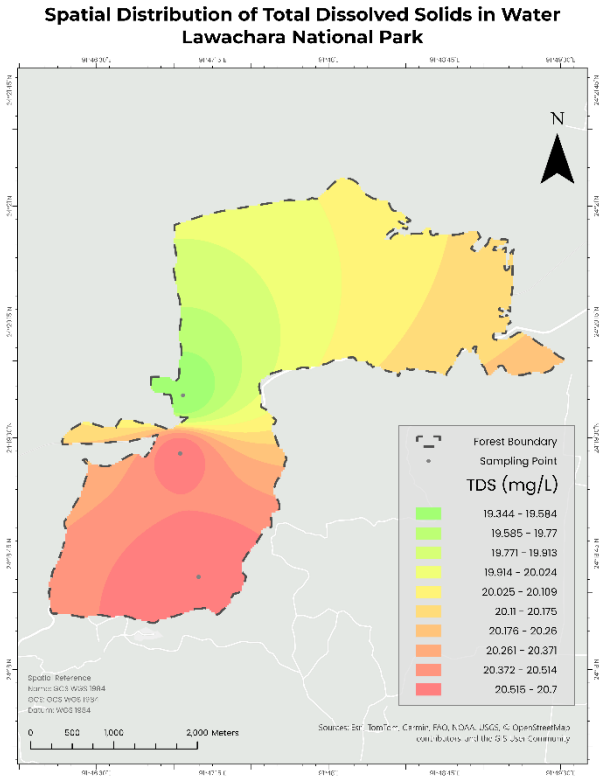


Figure 13 TDS, Lawachara

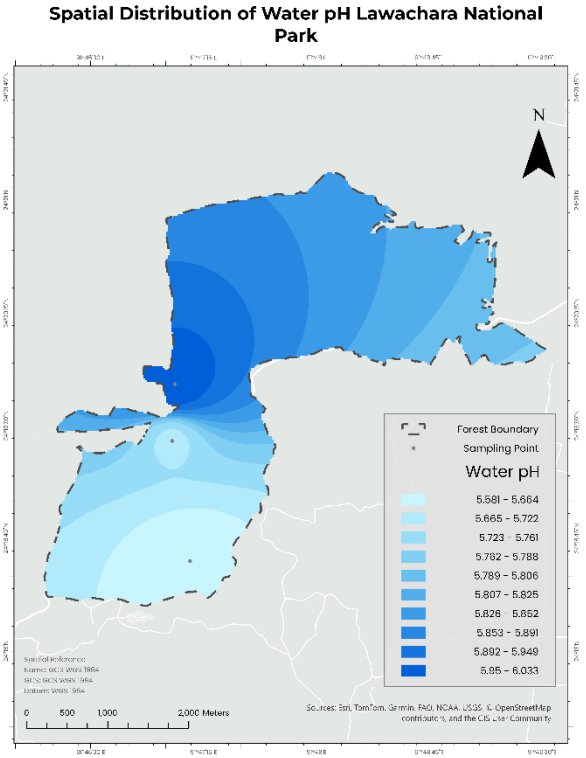


Figure 15 Water pH, Lawachara

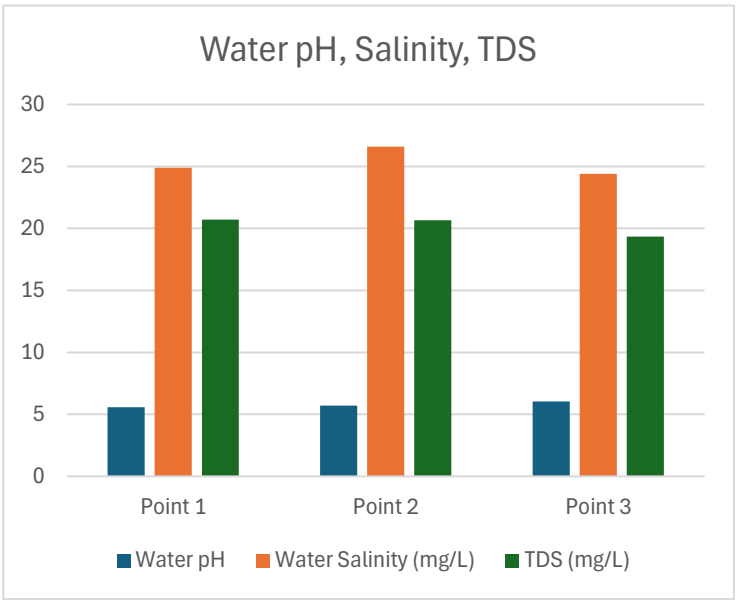


Figure 14 Water Quality Graph, Lawachara

Table 7 Water Quality, Lawachara

Location	S\N	Latitude	Longitude	TDS (mg/L)	Water Salinity (mg/L)	Water pH
Lawachara National Park	Point 1	24.31	91.786	20.7	24.9	5.58
	Point 2	24.32332	91.784013	20.67	26.6	5.7
	Point 3	24.329598	91.784321	19.343	24.4	6.033

Electrical Conductivity (EC) of Water:

Water at Lawachara National Park barely conducts electricity its EC hovers between 33.60 and 42.66 µS/cm at all three spots where samples were taken. That's way below the WHO's "excellent" threshold of 400 µS/cm, which means this water barely has any dissolved salts at all.

This low EC isn't just a number it matches up with really low levels of TDS and almost no salinity. Basically, these shaded forest streams are almost free of minerals, heavy metals, or chemical runoff from farms. It's the kind of pure water in a place where rainfall feeds the streams and the forest floor is left untouched. The thick trees and tangled roots do a solid job filtering out anything from the soil before water ever trickles into the streams. So, the water hasn't picked up much from the ground about as clean and unspoiled as it gets.

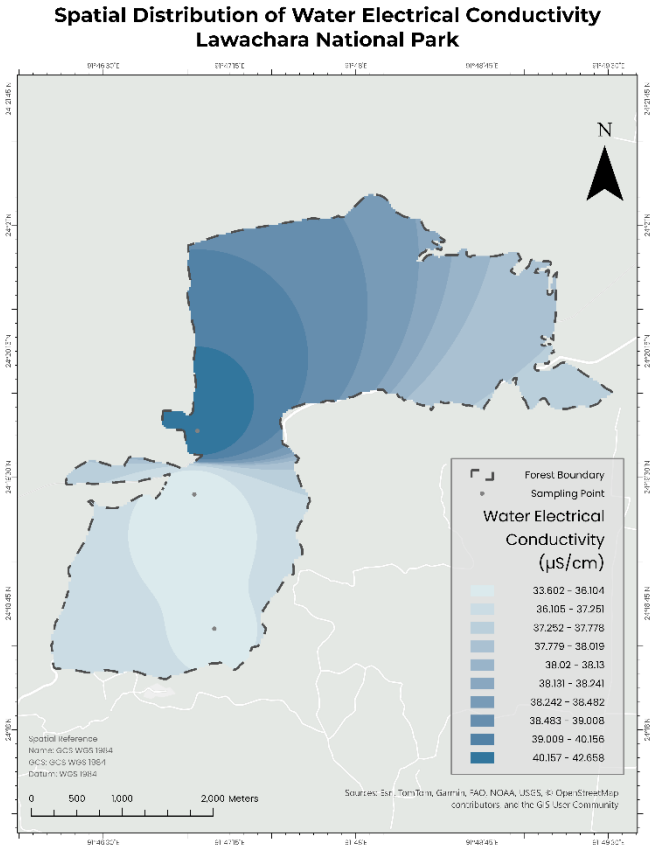


Figure 17 Water EC, Lawachara

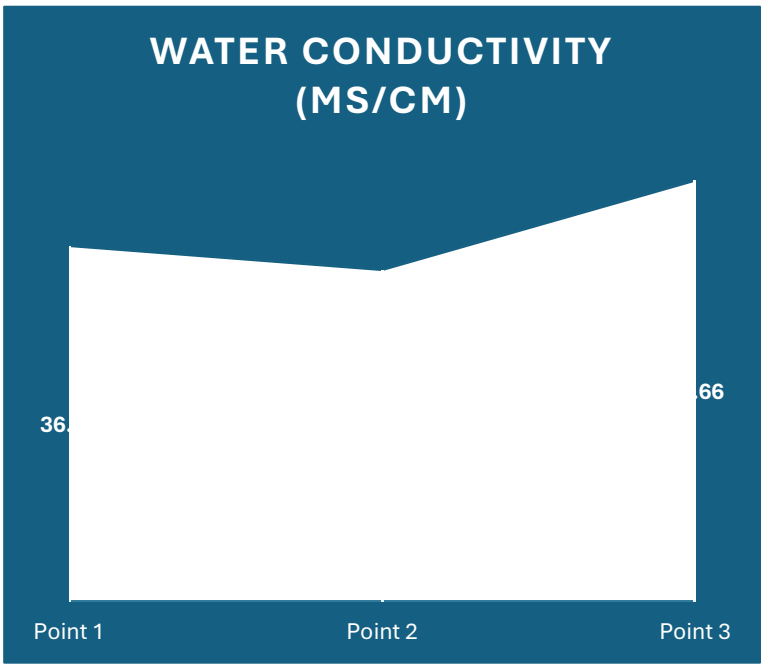


Figure 16 Water EC Graph, Lawachara

Table 8 Water EC, Lawachara

Location	S\N	Latitude	Longitude	Water Conductivity (µS/cm)	Avarage
Lawachara National Park	Point 1	24.31	91.786	36.03	37.43
	Point 2	24.32332	91.784013	33.6	
	Point 3	24.329598	91.784321	42.66	

Overall Water Condition:

Lawachara National Park's water is about as clean as it almost untouched, really. The numbers tell the story. By World Health Organization standards, there are barely any dissolved minerals or salts floating around. Total Dissolved Solids are under 21 mg/L, salinity sits below 27 mg/L, and electrical conductivity doesn't even hit 43 µS/cm. All of these are way below what's considered safe, making it clear there's nothing nasty no pollution, no heavy metals, and no runoff from farms.

Now, the pH runs a little low (between 5.58 and 6.03), which is just below the WHO's sweet spot for drinking water. But that's not a sign of anything bad. It's just nature doing its thing. The slightly acidic water comes from humic and fulvic acids leaching out of all the decaying leaves and organic stuff blanketing the forest floor. In the end, Lawachara's streams are pure, rain-fed, and filtered constantly by the thick forest soils. It's one of those rare places where the water really does live up to the "pristine" label.

Soil Condition

Soil pH and Moisture:

Soil pH at Lawachara National Park shows a lot of variation it ranges from 5.30 to 6.67 depending on where measured. Most environmental scientists and agricultural professionals consider soil pH between 6.0 and 7.5 ideal. But a reading like 5.3 isn't odd or challenging for a dense tropical forest. It simply means the soil is moderately acidic, which is perfectly usual and even good in conditions like these.

Leaves keep dropping in those thick forests, and honestly, it's a bigger deal than most people realize. As the leaves break down, they turn into things like humic compounds and fulvic acids. These acids seep into the ground and steadily boost the soil's acidity. That little change flips the whole environment, creating just the right setup for evergreen and semi-evergreen forests rich, acidic dirt full of organic matter and a steady stream of nutrients.

Soil moisture in Lawachara is seriously high anywhere from 28% to almost 47%. That's a lot of water, even for a tropical forest. For loamy or fertile ground, having anywhere between 20% and 50% is pretty much perfect for plants. Lawachara's soils stay this juicy for a couple of reasons. First, the dense canopy up above blocks a lot of direct sunlight, so the ground doesn't dry out as fast as it would in an open area. Then there's the thick layer of fallen leaves breaking down on the forest floor, which works like natural mulch. It locks in moisture right where plants need it most around their roots. Thanks to all this, the forest stays green and alive all through the year.

Soil Electrical Conductivity (EC):

The soil in Lawachara National Park showed a huge range of electrical conductivity, from just 35.33 µS/cm up to 202.67 µS/cm at the three sampling spots. That's way below the 2,000 µS/cm threshold, the cutoff for saline soils according to standard pedology and agriculture. So, it's safe to say the ground here is totally non-saline, with no worrisome salt buildup.

Actually, that's exactly what evergreen and semi-evergreen forest plants need, since they really don't handle salt stress well. The big difference in EC from spot to spot dipping as low as 35 then jumping past 200 just tells how it varied and lively the forest floor is. It gets pockets loaded with organic matter breaking down, scattered leaf litter, and constant nutrient cycling. Microbes are busy, decomposing things and releasing ions at different rates, all of which leads to those changing EC numbers in the topsoil.

Soil Texture:

The soil in Lawachara National Park, it's a patchwork. Using the ribbon length test, we saw some wild differences from spot to spot so the clay content and plasticity really bounce around. One sample gave us a ribbon length of 0.0 mm, which means the soil couldn't even stick together; that's typical loamy sand, packed with leaf litter instead of clay minerals. At another point, it was 17.33 mm more like a balanced loam, with enough clay for structure and decent drainage. Then, out of nowhere, 69.33 mm a thick, sticky ribbon that screams "heavy clay" with strong water retention. The soil's constantly shifting, thanks to changes in micro-topography, how the organic matter breaks down, and the geology beneath the surface. So, in just a few steps, moving from loose, crumbly, organic-rich topsoil to tough, dense clay patches. It's chaotic, but that's exactly what can be found in a natural, untouched forest floor.

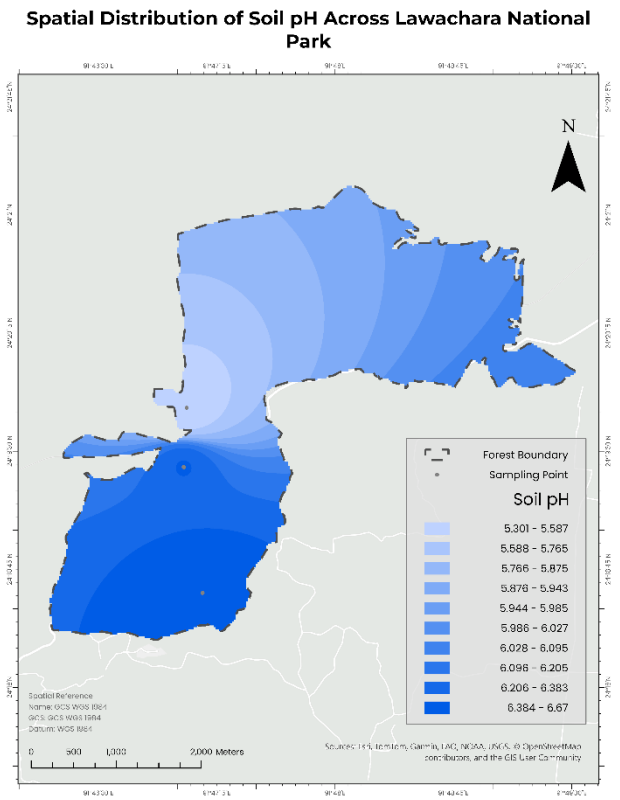


Figure 20 Soil pH, Lawachara

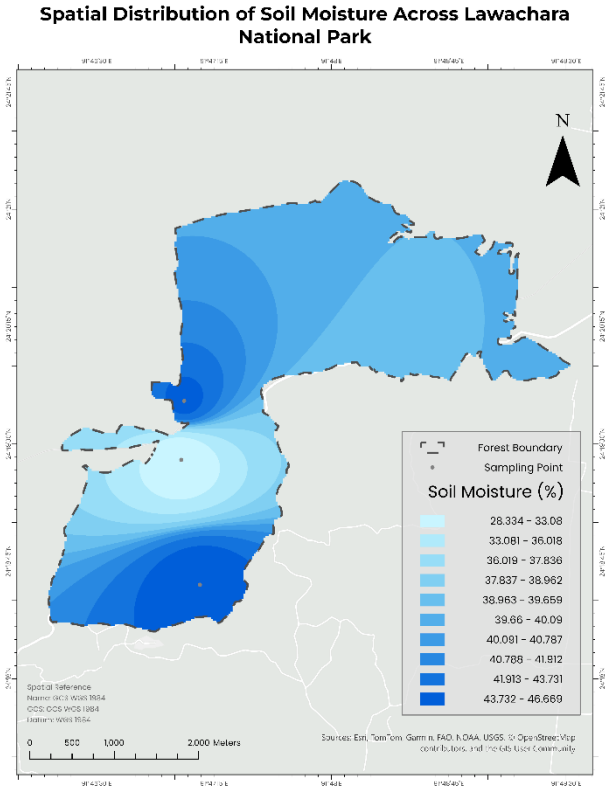


Figure 21 Soil Moisture, Lawachara

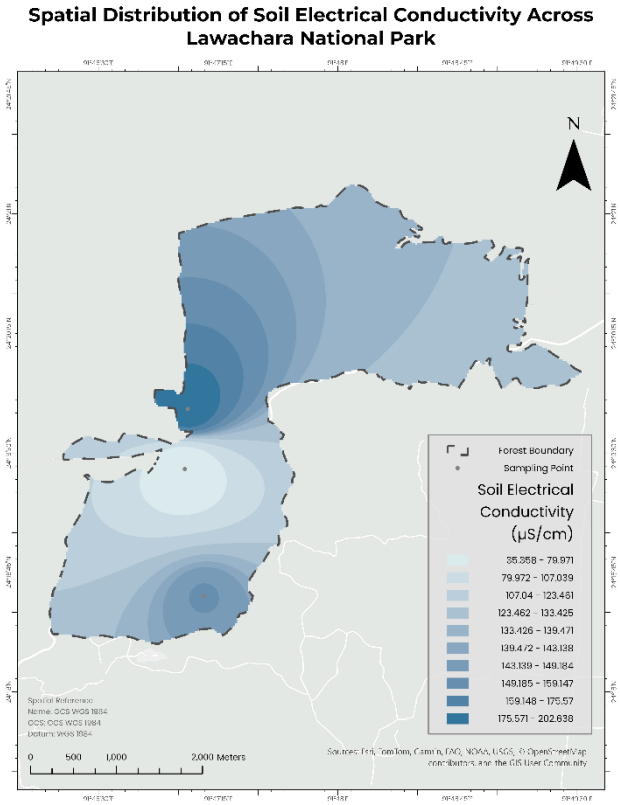


Figure 19 Soil EC, Lawachara

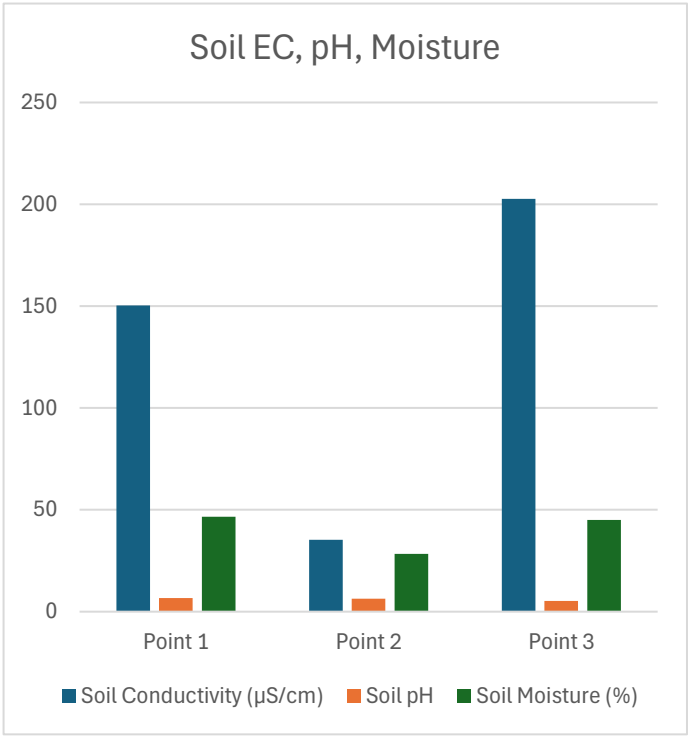


Figure 18 Soil EC,pH,Moisture Graph, Lawachara

Table 9 Soil EC,pH,Moisture, Lawachara

Location	S\N	Latitude	Longitude	Soil Conductivity (µS/cm)	Soil pH	Soil Moisture (%)	Soil Ribon (mm)
Lawachara National Park	Point 1	24.31	91.786	150.33	6.67	46.67	0
	Point 2	24.32332	91.784013	35.33	6.4	28.33	17.33
	Point 3	24.329598	91.784321	202.667	5.3	45	69.33

Overall Soil Condition:

Lawachara National Park has soil that's hard to beat deep, fertile, and loaded with organic matter. This soil in this spot isn't just good; it's almost perfectly suited for a tropical evergreen forest. The soil's rich, dark profile, hardly ever disturbed and always full of life. The pH level sits comfortably between 5.3 and 6.67, that is ideal for woodland growth. That's mostly because leaves and other plant debris break down constantly, piling up as a thick, nutritious mulch. This decaying layer not only feeds the soil; it keeps the moisture locked in between 28% and 47% so the ground stays damp, despite sunny weather.

One thing really stands out: the soil's electrical conductivity is super low, hugging the range of 35 to 203 µS/cm. That means it's free from salt and unwanted minerals. Plants absolutely thrive in these gentle conditions. And the texture of the soil well, it changes fast. A few spots are soft and loamy, buzzing with organics, but some are clay-dense, holding onto water and providing root support. This variety helps keep nutrients moving and gives plants something solid to anchor to.

Lawachara's soil is almost the perfect setting for a wild, dense forest. The roots get plenty of air and water, the ground stays strong, and everything above flourishes. It's a textbook example of nature working exactly as it should, fueling that tangled green world from below.

Overall Environmental Condition:

Lawachara National Park feels untouched like a perfect snapshot of a tropical evergreen forest doing exactly what it's supposed to do. The place is steady, healthy, and surprisingly undisturbed. For a model protected reserve, this forest fits the bill.

The canopy overhead isn't just for show, either. It does the heavy lifting for the whole ecosystem, keeping things in check. It acts like a thick blanket against heat and wind. The forest is calm, cool air averaging around 29°C even when it's hot outside. There's very little wind, and the humidity is very

elevated, often hitting 82%. That thick green canopy isn't just for show it's like nature's own air purifier, filtering out dust and all those tiny particles so the air always feels crisp and clean.

In Lawachara, everything stays in balance. Water and soil really suit the local plants. The streams flow crystal clear, carrying water that's almost pure practically free of minerals and salt. The numbers back it up: TDS stays under 21 mg/L, and electrical conductivity is below 43 µS/cm. The water and soil both tend toward the acidic side (water pH hovers around 5.6 to 6, soil pH lands between 5.3 and 6.7), which isn't surprising when its about all the leaf litter constantly breaking down out there. This layer of decaying leaves does wonders: it holds onto water, keeping soil moist at times reaching 47% and creates rich, fertile ground varying from loose, organic loam to moist clay.

Because it's mostly been spared from over-harvesting and pollution that trouble other areas nearby, Lawachara still thrives. The ecosystem stays balanced, well-nourished, and wrapped in a kind of natural insulation, letting the park stay healthy and wild.

4.2.2. Jaflong Zero Point

Micro-meteorological Condition

Temperature:

Jaflong is noticeably warmer there's no getting around it. During sampling, surface temperatures hovered between 34.13°C and 35.27°C, never really dropping. Unlike the woodland reserves, where the forest canopy acts like a shield, here the open land just soaks up sunlight all day. With nothing to block the sun, heat piles up fast, making the air heavy and the ground hotter. The result? Jaflong's microclimate feels harsher and far less forgiving than the cool, shaded spots found elsewhere in Sylhet.

Humidity:

Jaflong feels noticeably drier. The relative humidity drops to between 43% and 51%, depending on spot. With so much open land and sunlight blasting the area, any surface moisture disappears quickly. There's no dense cover of trees or plants holding onto water and slowly releasing it into the air. The dryness isn't subtle it's clear that the lack of vegetation and full exposure to the sun alter the atmosphere entirely. The moment was a world apart from the thick, humid air of a shaded forest.

Atmospheric Pressure:

Atmospheric pressure at Jaflong barely budged during the study, staying between 1005.53 and 1008.20 hPa. Jaflong isn't wrapped up like a thickly wooded forest instead, it faces the elements head-on, which usually means weather patterns can have more direct effects. Still, the pressure hardly changed, signaling that overall synoptic conditions stayed calm even in this open, unprotected setting.

Table 10 Atmosphere Condition, Jaflong

Spot	S\N	Lat	Long	Temperature (C)	RH (%)	Pressure (atm)	WS (m/s)	WD
Jaflong	Point 1	25.18395	92.00957	34.63	50.6	1008.2	1.07	75.3
	Point 2	25.1829	92.0172	35.27	43.33	1006.26	0.9	173.33
	Point 3	25.18596	92.0057	34.13	44.3	1005.53	0.5	96.67

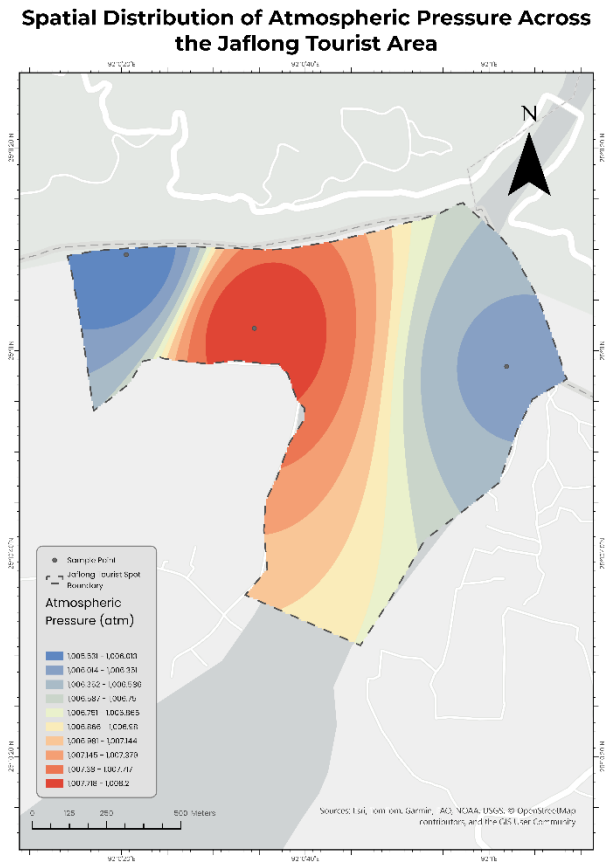


Figure 24 Presure, Jaflong

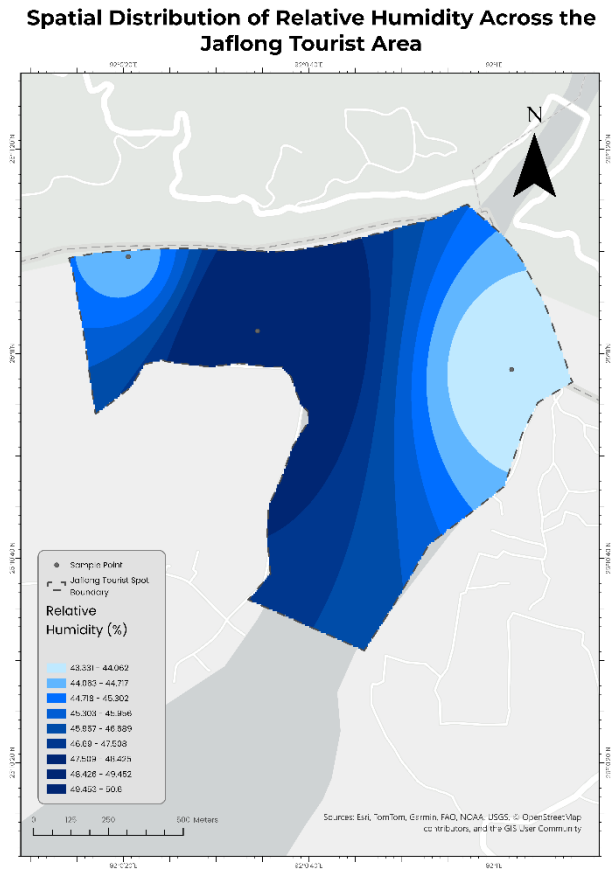


Figure 25 RH, Jaflong

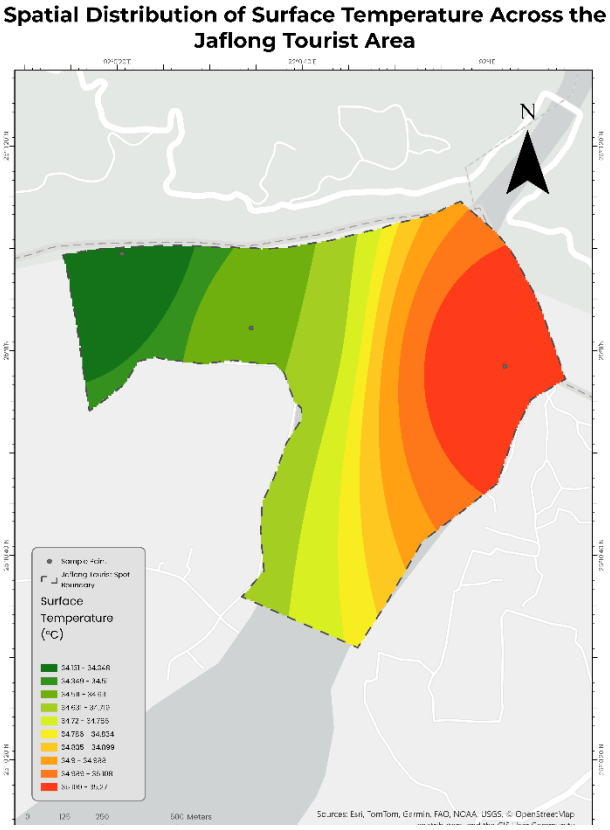


Figure 23 Temperature, Jaflong

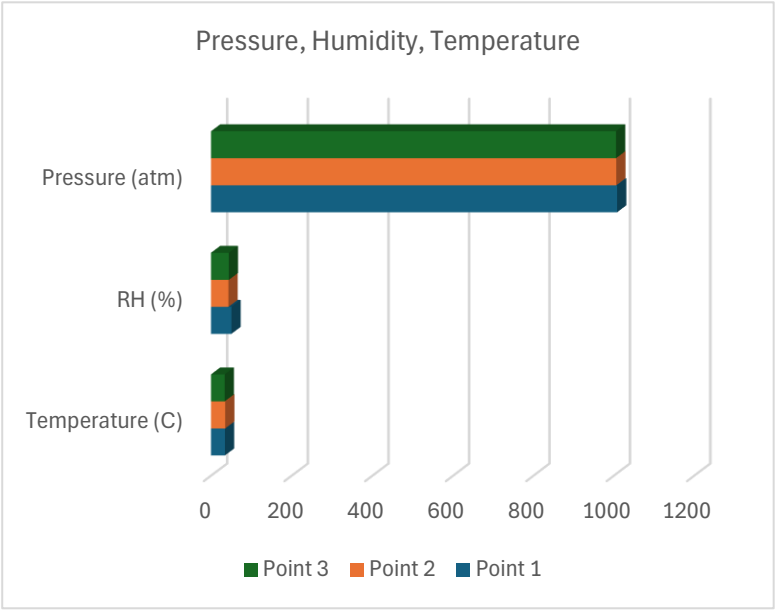


Figure 22 Pressure, RH, Temp., Jaflong

Wind Speed and Direction:

In Jaflong, the air moving. Wind speeds ranged from 0.50 to 1.07 m/s, and the direction shifted between 75.30° and 173.33°. Without thick forests or tall tree lines to block it, the wind just rolls through the open land. That steady breeze makes the heat more bearable. The air never feels completely still it keeps the place lively and gives the landscape a constant, refreshing energy.

Overall Weather Condition:

Jaflong packs a punch it's hot, wide open, and wind sweeps through like a constant companion. The temperature hits 35.2°C, and the humidity sticks in the 40s, so the air feels dry. Sunlight dominates, with nothing around to provide shade. The breeze doesn't let up, running up to 1.07 m/s, and stirs things just enough to keep the place from feeling oppressive. Pressure stays level, but everything here feels alive and exposed. This landscape bakes in the sun, a sharp break from the usual thick, humid forests that feel closed in and sheltered.

Air Quality

Particulate Matter (PM 2.5 and PM 10):

Jaflong's air is impressively clean. When measured PM 2.5 was really fine, hard-to-see stuff it barely even shows up, hanging around just 3.00 to 4.33 micrograms per cubic meter. Even the bigger particles, PM 10, don't climb past 5.67 μg/m³. So heavy dust or haze was not hanging in the air. The steady wind helps a lot here; it sweeps through the open landscape, scattering any dust before it can settle or build up. This keeps the air fresh and remarkably clear.

Table 11 Air Quality

Spot	S\N	Lat	Long	PM 2.5	PM 10
Jaflong	Point 1	25.18395	92.00957	4.33	5.33
	Point 2	25.1829	92.0172	3	4
	Point 3	25.18596	92.0057	3.33	5.67

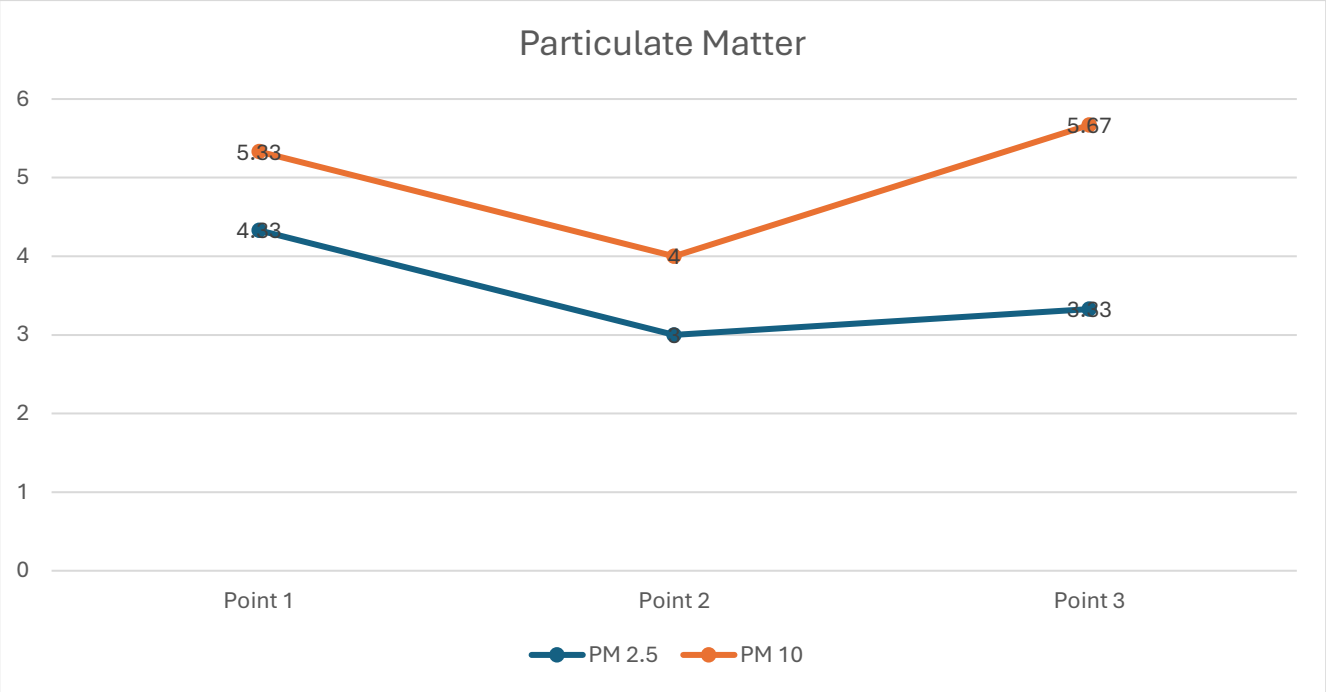


Figure 28 PM Graph, Jaflong

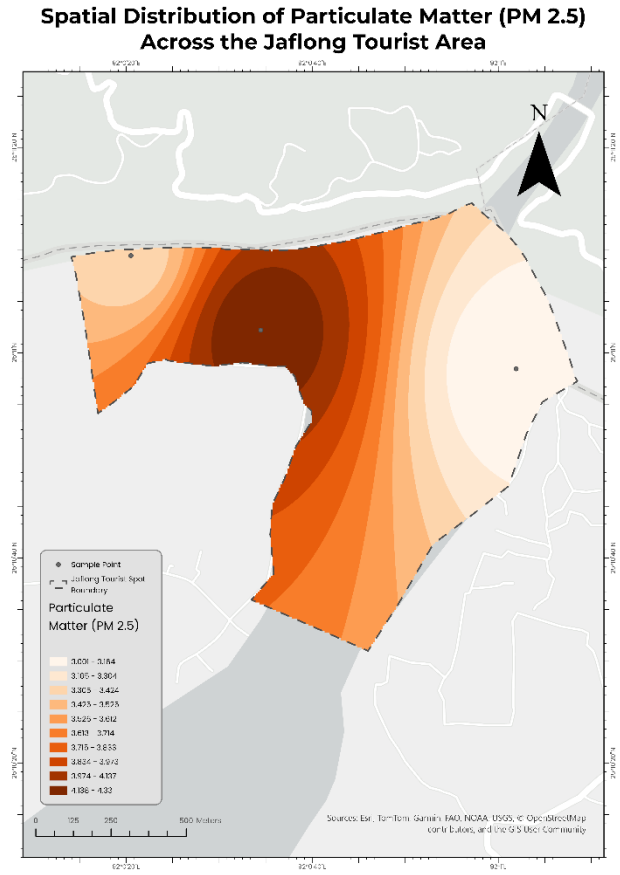


Figure 26 PM 2.5, Jaflong

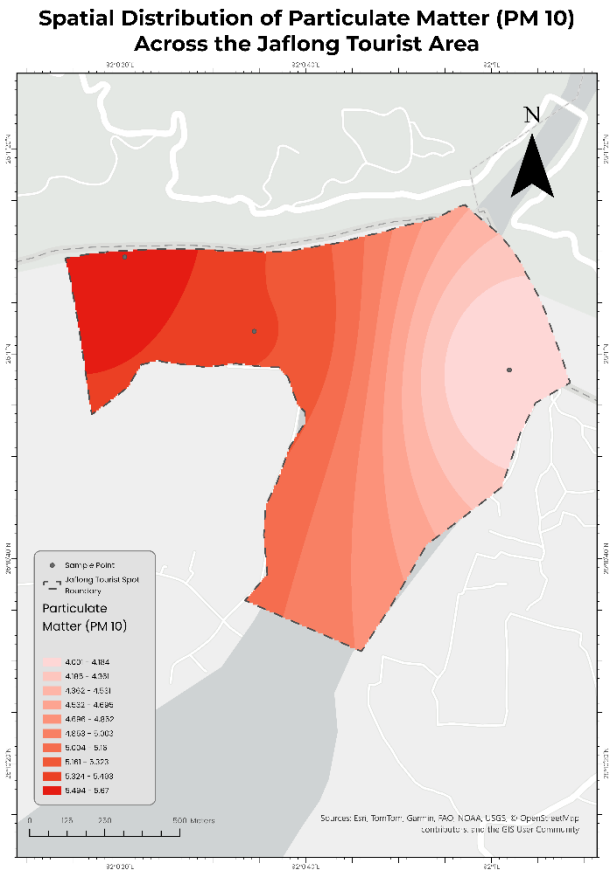


Figure 27 PM 10, Jaflong

Overall Air Quality:

Jaflong stands out for its incredibly clean air. Even though the area is open, there was not much particulate matter floating around it's just not an issue here. The steady breezes play a big part, always moving and clearing the air before anything can settle. Because of this, the whole place feels fresh and invigorating, with air that really lives up to the highest standards.

Water Quality

Total Dissolved Solids (TDS):

Jaflong's water looks impressively clean. Total Dissolved Solids (TDS) levels sit between 39.96 and 115.20 mg/L. That's great consideration the World Health Organization sets anything below 300 mg/L as excellent for drinking. Sure, it isn't quite as pristine as some deep forest streams, but it's close remarkably pure, barely any suspended sediment or heavy minerals. The local hydrology stays clear, hardly touched by dissolved contaminants.

Water Salinity:

At Jaflong, water salinity stays low right between 27.30 mg/L and 75.33 mg/L across all three sampling points. That's nowhere near the World Health Organization's limit of 500 mg/L for top-notch freshwater. These streams are genuinely fresh. Salt is not leaching out from rocks, or pollution seeping in from factories. The water looks clean, healthy, and naturally renewed.

Water pH:

Water at Jaflong strikes just the right balance. Tests put the pH between 7.36 and 7.45, well within the World Health Organization's safe range for drinking 6.5 to 8.5. This isn't like the acidic streams in dense forests; it's slightly alkaline, which is typical for open rivers that pick up minerals from rocks and soil. That subtle alkalinity points to a well-balanced, healthy ecosystem exactly what fish and other local species need to thrive.

Electrical Conductivity (EC) of Water:

Water at Jaflong barely conducts electricity EC measurements range from 37.50 to 115.47 µS/cm in all locations. That's well under the WHO's "excellent" cutoff of 400 µS/cm, so there aren't many dissolved inorganic salts. The low EC fits right in with the great TDS and salinity readings. In short, this isn't water polluted with heavy metals or loaded with chemicals. It's clean and only lightly mineralized, thanks to the way the local landscape naturally refreshes it.

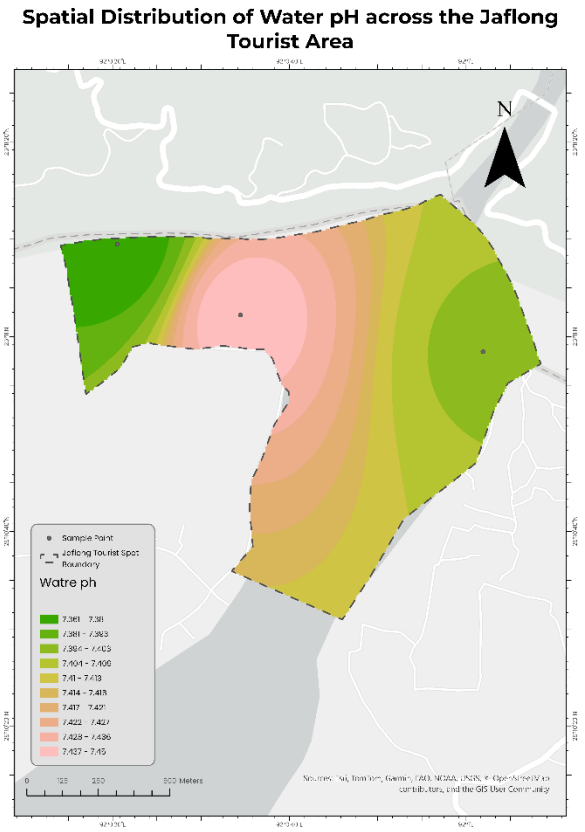


Figure 31 Water pH, Jaflong

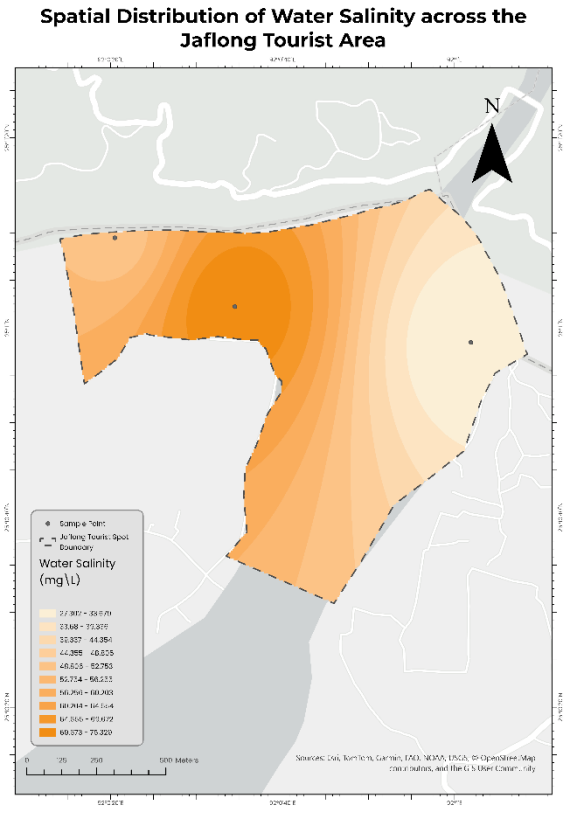


Figure 32 water Salinity, Jaflong

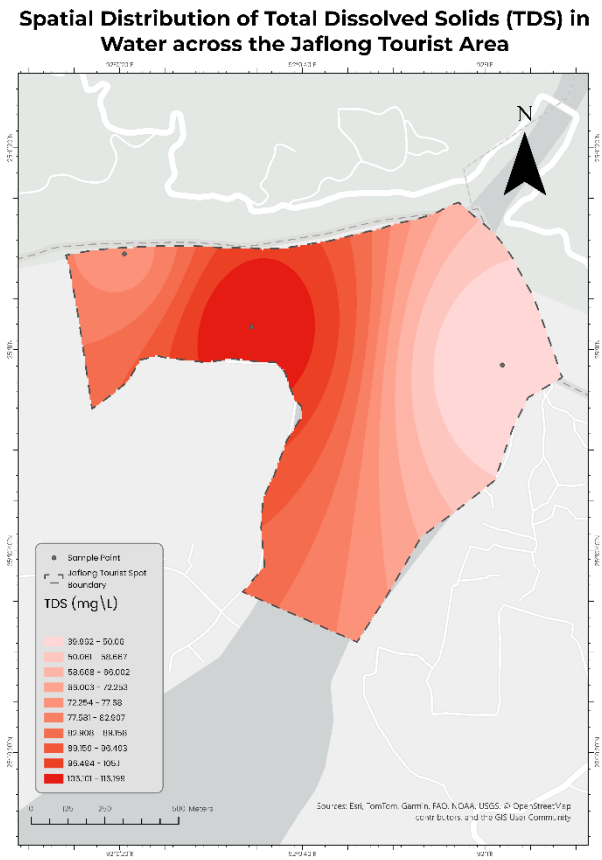


Figure 30 TDS, Jaflong

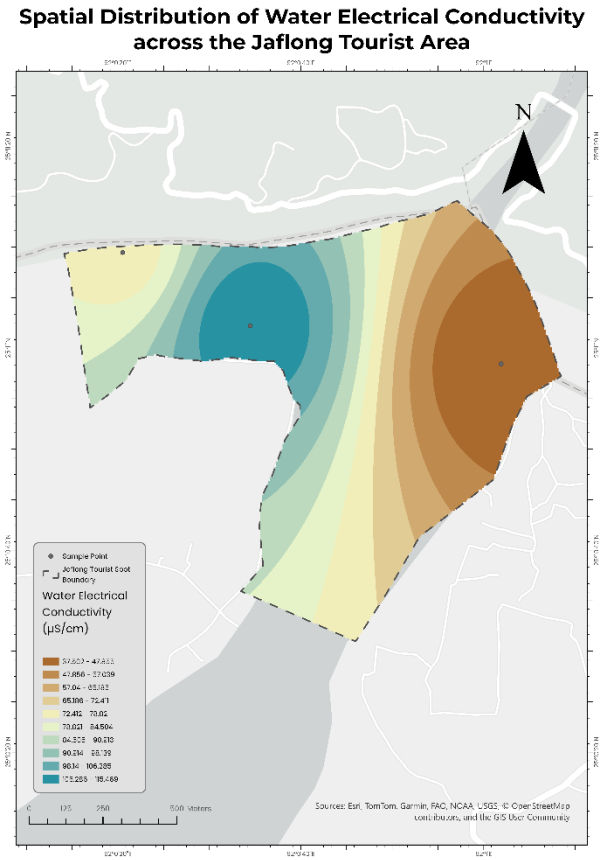


Figure 29 Water EC, Jaflong

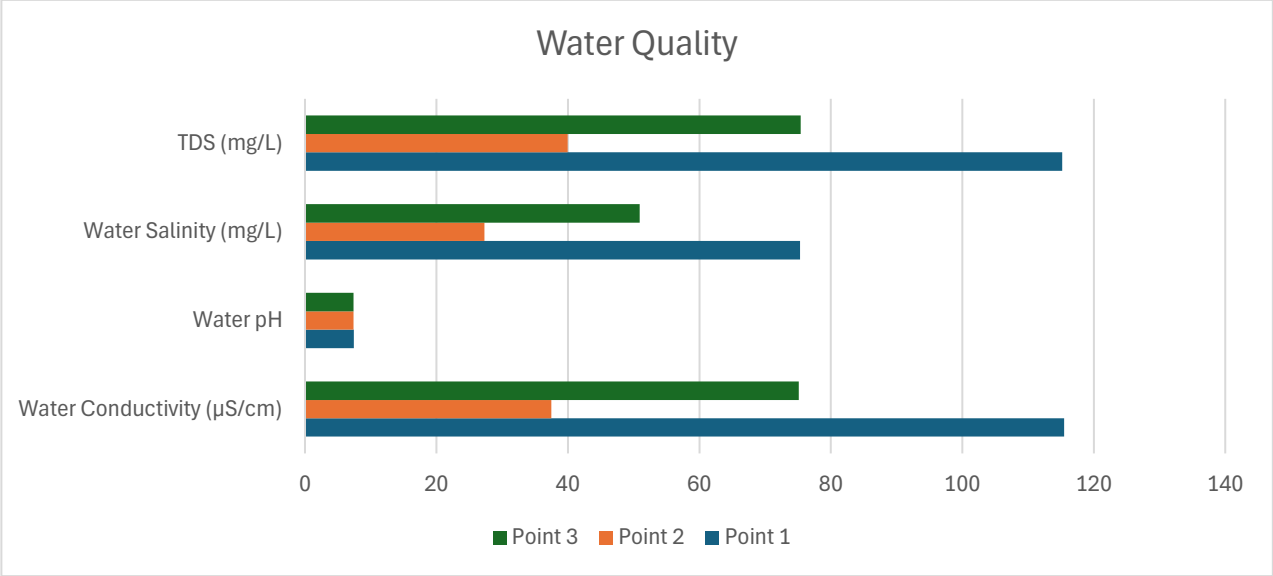


Figure 33 Water Quality Graph, Jaflong

Table 12 Water Quality, Jaflong

Spot	S\N	Lat	Long	Water Conductivity (µS/cm)	Water pH	Water Salinity (mg/L)	TDS (mg/L)
Jaflong	Point 1	25.18395	92.00957	115.47	7.45	75.33	115.2
	Point 2	25.1829	92.0172	37.5	7.4	27.3	39.96
	Point 3	25.18596	92.0057	75.13	7.36	50.93	75.4

Overall Water Condition:

Jaflong's water really stands out it's clean, healthy, and balanced. According to World Health Organization standards, this water ranks at the top. Total Dissolved Solids hover around 115 mg/L, salinity stays well under 76 mg/L, and electrical conductivity doesn't go beyond 116 µS/cm. These numbers show there's no heavy pollution or dangerous runoff messing things up. The pH impresses too, sitting comfortably between 7.36 and 7.45 right inside the optimal WHO range. That's a shade alkaline, exactly expected from a vibrant river system. All in all, Jaflong's water truly delivers: clean, clear, and beautifully balanced.

Soil Condition

Soil pH and Moisture:

At Jaflong, soil pH swings from moderately acidic to almost neutral anywhere between 5.40 and 6.67, depending on the spot. That's a sweet spot for most native plants; they thrive in this kind of soil. Moisture levels vary quite a bit, too, from as low as 23.33% up to 43.33%. Even when the sun beats down and the air feels bone-dry, the soil keeps a good amount of water. What's happening beneath the surface matters here. The groundwater and those regular floods from the river recharge the soil, so the ground stays hydrated. Despite hot days and dry air, the soil keeps delivering what plants need.

Soil Electrical Conductivity (EC):

Jaflong's soil has electrical conductivity between 18.67 and 169.00 µS/cm, way below the 2,000 µS/cm mark for saline soils. So, the salt levels aren't a problem this ground is genuinely non-saline. Still, the changes in EC tell us there's a lot happening underground. Nutrients keep moving, minerals shift around, and the soil stays lively and healthy.

Soil Texture:

Jaflong's soil feels loose underfoot and water just slips right through it. In the ribbon length test, results stretched from 0.0 mm up to 23.67 mm. When the reading hits zero, soil was pure sand crumbly stuff that won't hold together, which usually find along riverbanks and areas packed with fresh sediments. At the upper end, 23.67 mm signals a sandy loam or maybe a balanced loam, where there's enough clay mixed in to hold a bit of water without turning into a heavy, sticky mess. There can be found thick clay patches here. Instead, the ground is a quick-draining blend of light loam and sand easy to work with, but holding onto water isn't its strength.

Table 13 Soil Quality, Jaflong

Spot	S\N	Lat	Long	Soil Conductivity (µS/cm)	Soil pH	Soil Moisture (%)	Soil Ribon (mm)
Jaflong	Point 1	25.18395	92.00957	169	5.83	36.37	4
	Point 2	25.1829	92.0172	18.67	5.4	43.33	23.67
	Point 3	25.18596	92.0057	46.83	6.67	23.33	0

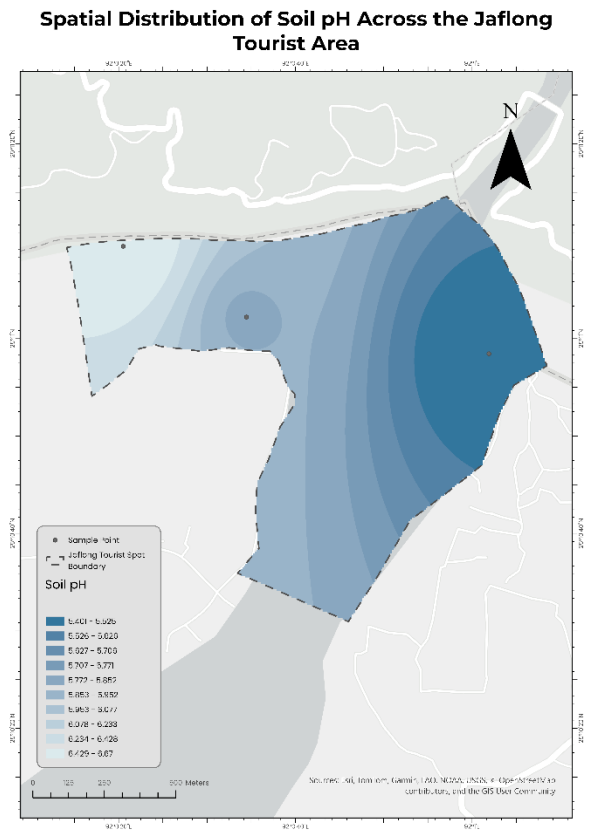


Figure 34 Soil pH, Jaflong

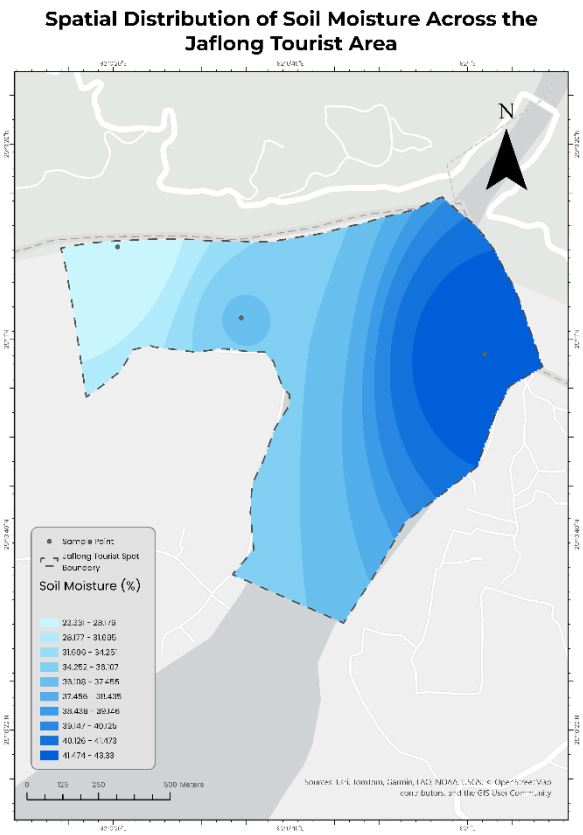


Figure 35 Soil Moisture, Jaflong

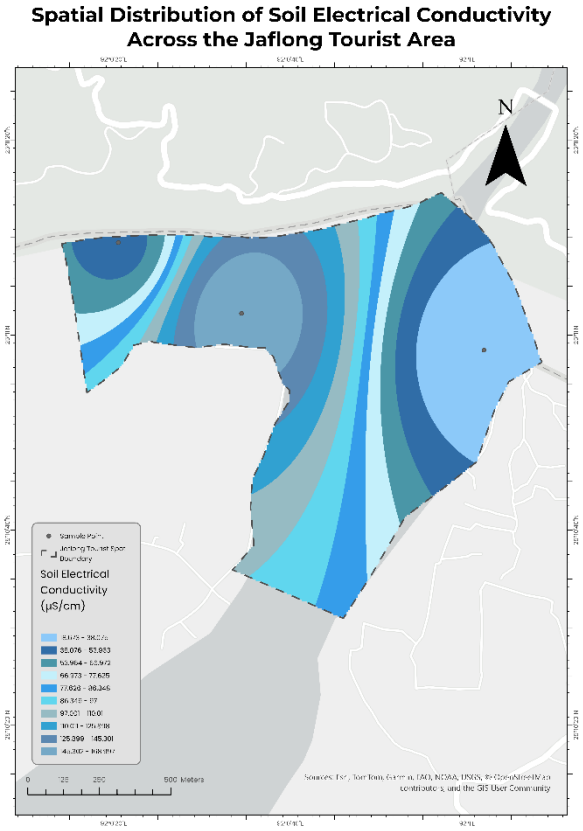


Figure 36 Soil EC, Jaflong

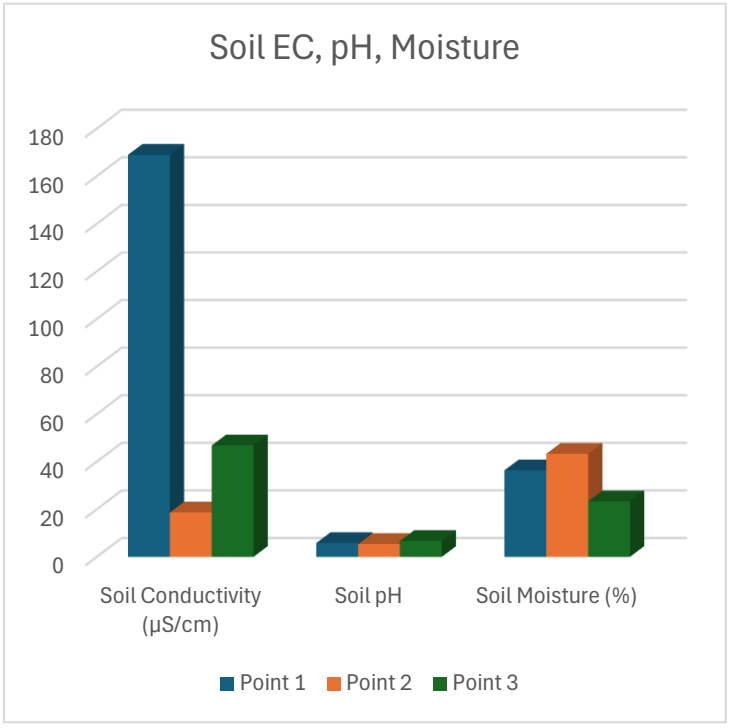


Figure 37 Soil Quality Graph, Jaflong

Overall Soil Condition:

Jaflong's soil tells a story shaped by the nearby rivers. It's lively never stagnant and everything about it supports plants that call this place home. Check the pH: it floats between 5.40 and 6.67. That range sets up the local greenery for healthy growth. Even when the surface bakes in the sun, the soil hangs on to moisture usually somewhere from 23% to 43%. So, the ecosystem rarely dries out.

On top of that, the soil barely conducts electricity under 169 µS/cm so there's no salt stress troubling the plants. There wasn't find thick, heavy clay. Instead, it's all sand and loam light, crumbly, and quick to drain, unlike the waterlogged soils hidden deeper in the forests. These loose soils let roots stretch out and breathe, but they still keep enough water for plants to power through the hottest days. Jaflong's landscape isn't just beautiful, it's built for life.

Overall Environmental Condition:

Jaflong doesn't play it subtle it's bold, hot, and startlingly clean. Unlike the dim coolness under a forest canopy, everything here sparkles under hard sunlight. The temperature soars above 34°C, and the air holds its dryness, sticking to humidity between 43% and 50%. Still, the wind doesn't let things get uncomfortable. It flies through the open space, stirring everything up and keeping the air lively. These winds do more than just cool sweep dust and pollutants away. That's why particulate matter stays low, so breathe is crisp, clear air without grit or haze.

Water here is as pure as it gets. Test it and found pH between 7.36 and 7.45 the sweet spot set by the WHO for drinking. It's soft and clean, carrying hardly any dissolved minerals (TDS less than 116 mg/L) and almost no salt. It is top-tier, alkaline freshwater, Jaflong delivers. The ground underfoot tells the same story. The soil isn't salty, and even with the heat, it stays moist enough. It's loose, sandy loam excellent for draining away rain, so nothing gets muddy or waterlogged.

In the end, Jaflong's ecosystem stands in stark contrast to closed forests. There's no shade, but the environment thrives in its own way bathed in sun, swept by pure air, supported by pristine water and soil. It's a landscape built for resilience, health, and life in the open.

4.2.3 Sada Pathor, Bholaganj

Micro-meteorological Condition

Temperature:

Temperature at Sada Pathor (Bholaganj) is brutal. During the sampling period, the surface remained scorching temperatures floated between 40.20°C and 40.96°C and didn't let up. No shade, no relief. The whole place is covered in glaring white rocks and wide-open riverbeds. Sunlight just bounces off every surface, amplifying the heat. With no forest canopy overhead, the land doesn't have anything no insulation, no cooling from leaves to soften those fierce temperatures. Heat builds up fast here, so the microclimate ends up much hotter and harsher than any shaded, jungle-like area in Sylhet.

Humidity:

Humidity isn't forgiving either. It registers as moderate but feels suffocating relative humidity falls between 58.45% and 63.00% in the sampling locations. The open exposure means moisture doesn't stick around; evaporation works overtime. Yet the nearby river keeps a constant flow of water vapor in the air, so it never feels dry. The mix of relentless heat and this steady humidity creates a muggy, oppressive atmosphere. Without thick vegetation to balance the moisture, the air feels harsher and far more uncomfortable than what experience deep in the moisture-rich forests.

Atmospheric Pressure:

Atmospheric pressure at Sada Pathor stayed almost unchanged throughout the sampling just minor fluctuations from 1003.1 to 1003.6 hPa. The weather behind these numbers didn't budge; everything stayed steady and quiet. Even though Sada Pathor stands wide open along the river, synoptic conditions refused to stir. This kind of unwavering, low pressure fits the place: thick, relentless heat settles in and barely moves. It created a dense, stagnant microclimate while researchers observed.

Wind Speed and Direction:

At Sada Pathor, nothing moved not even the air. Wind speed hit zero everywhere we checked, so tracking direction was out of the question. The place is wide open and rocky, but the complete absence of a breeze only ramps up the heat. With no wind to carry it off, the white stones just throw that warmth right back. The atmosphere sits heavy, the pressure stays the same, and the hot, slightly humid air hangs there, trapped. It's a stagnant, brutally intense spot heat collecting, never leaving.

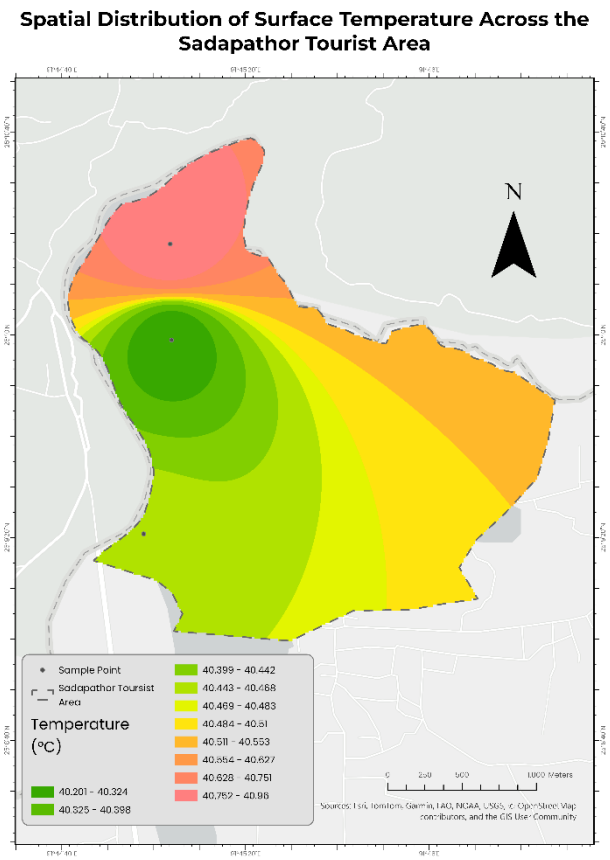


Figure 41 Temp., Sada Pathor

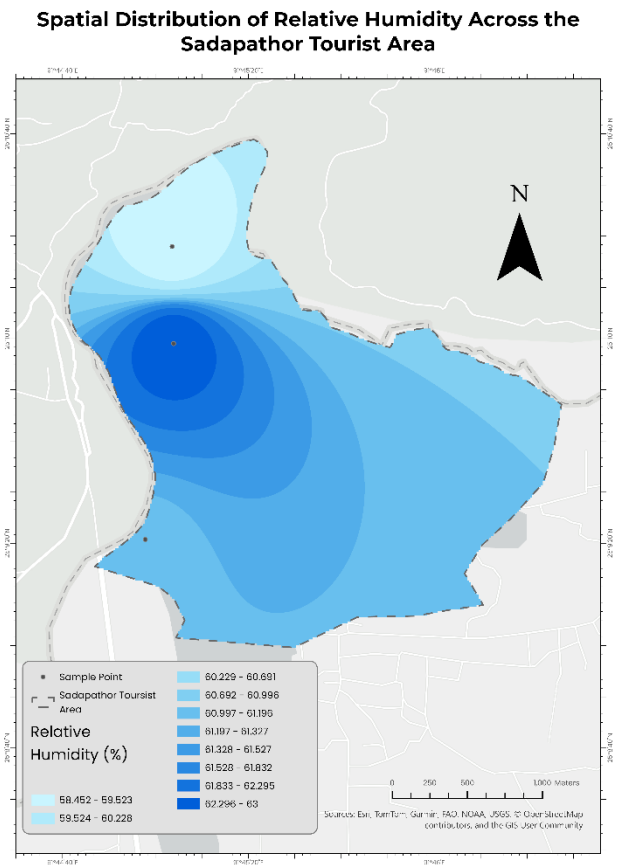


Figure 40 RH, Sada Pathor

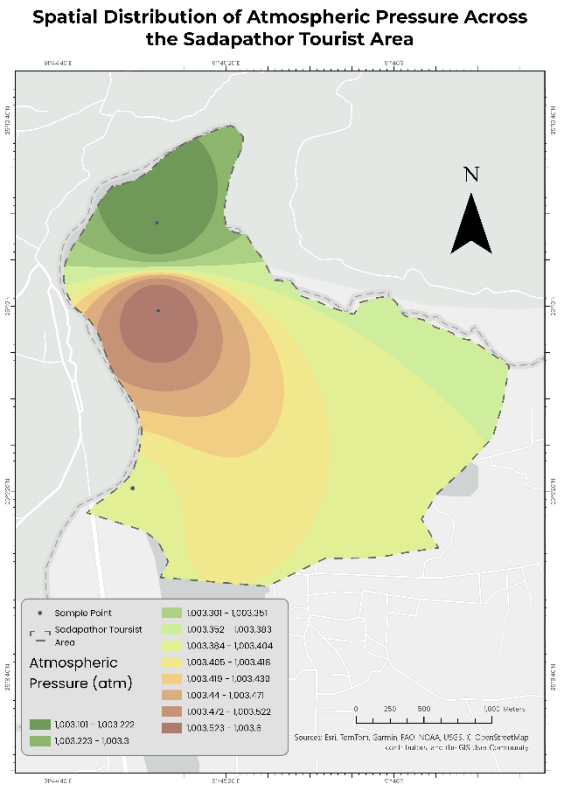


Figure 38 Pressure, Sada Pathor

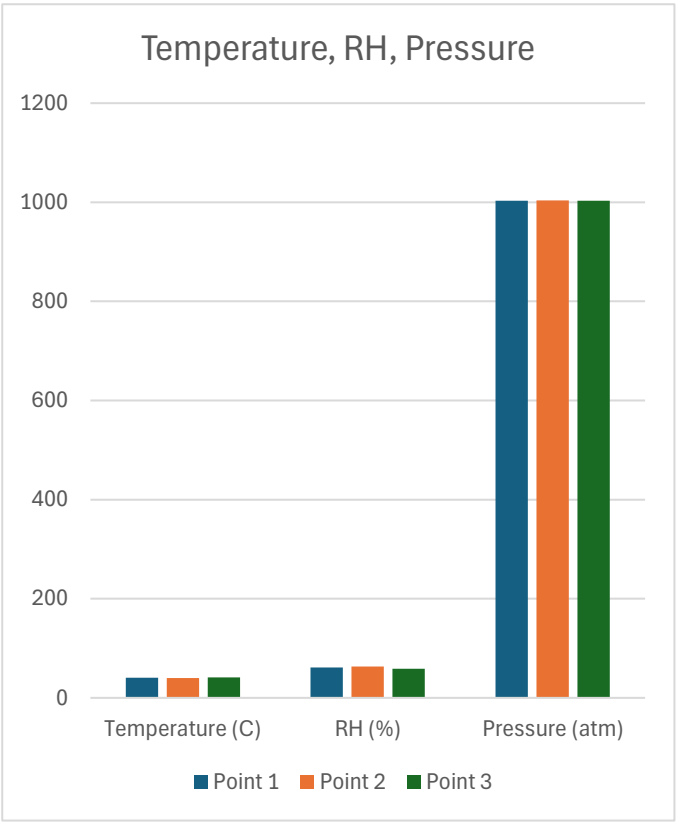


Figure 39 Temp. RH, Pressure, Sada Pathor

Table 14 Atmosphere Condition, Sada Pathor

Place	S\N	Lat	Long	Temperature (C)	RH (%)	Pressure (atm)	WS (m/s)	WD
Sada Pathor	Point 1	25.15576	91.7494	40.46	61.13	1003.4	NA	NA
	Point 2	25.16639	91.75109	40.2	63	1003.6	NA	NA
	Point 3	25.17166	91.75101	40.96	58.45	1003.1	NA	NA

Overall Weather Condition:

Sada Pathor doesn't just get hot it turns brutal. At outside and it's like walking straight into a solar oven. The mercury doesn't bother stopping at 40°C, and when the 60% humidity, the air turns thick, almost suffocating. But what really grinds down is the silence of the air. Not a single breath of wind stirs. The heat just hangs there, trapped above the rocks, baking everything in sight. The atmospheric pressure hardly budges, so nothing moves everything stays locked in this heavy, sun-soaked stillness. And it all comes down to what is seen overhead: nothing but sky. No canopy, nothing to break the glare, just rocky ground throwing sunlight right back. In the end, what is a microclimate that's fierce, raw, and leaves nowhere to hide.

Air Quality

Particulate Matter (PM 2.5 and PM 10):

Particulate matter levels at Sada Pathor show the direct impact of its exposed, active landscape. PM 2.5 values sit noticeably higher than in protected forests, ranging between 12.33 and 18.00 µg/m³ at the different sampling spots. The larger particles, measured as PM 10, range between 14.33 and 22.67 µg/m³. With numbers like these, it's clear there is a moderate amount of dust and suspended aerosols floating around. The total lack of a forest canopy means there are no leaves to catch and filter out this airborne dust, which is likely kicked up by local geological activity, dry winds from nearby areas, or the lack of vegetative ground cover.

Overall Air Quality:

Sada Pathor has acceptable but noticeably dustier air quality compared to enclosed reserves. With PM 2.5 and PM 10 levels reaching up to 18.00 and 22.67 µg/m³ respectively, particulate pollution is present but not severely hazardous. The complete lack of trees means there is no natural air filter to trap these particles. Because the air here is stagnant (0 m/s wind), the dust that does get kicked up tends to hang in the atmosphere rather than blowing away, resulting in an environment that, while natural, lacks the pure, undisturbed air quality of a dense forest.

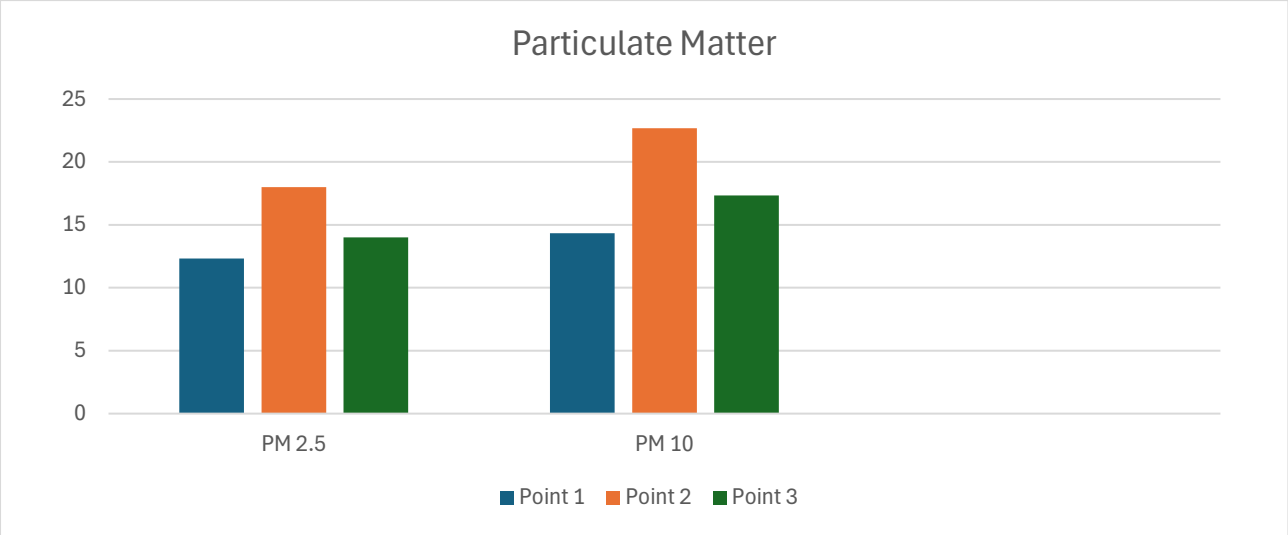


Figure 42 PM Graph, Sada Pathor

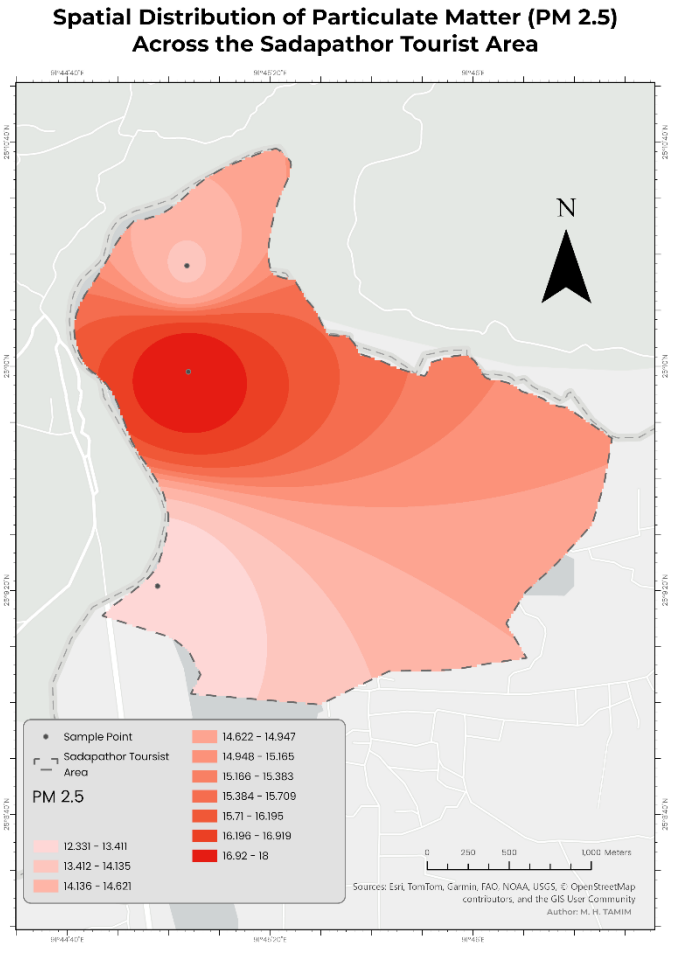


Figure 43PM 2.5, Sada Pathor

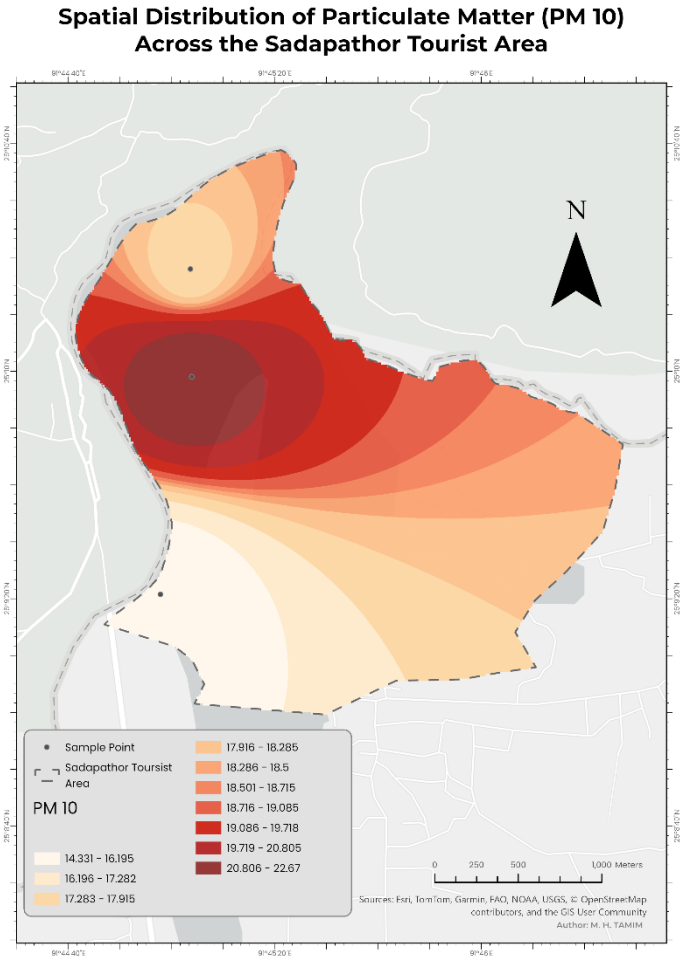


Figure 44 PM 10, Sada Pathor

Place	S\N	Lat	Long	PM 2.5	PM 10
Sada Pathor	Point 1	25.15576	91.7494	12.33	14.33
	Point 2	25.16639	91.75109	18	22.67
	Point 3	25.17166	91.75101	14	17.33

Water Quality

Total Dissolved Solids (TDS):

Sada Pathor's river water is impressively pure. In the sample water here, the Total Dissolved Solids (TDS) remains very low, fluctuating between 19.53 and 56.43 mg/L. This is exceptionally good. For reference, the World Health Organization rates anything under 300 mg/L as excellent for drinking. The lower readings at Sada Pathor are practically untouched, containing barely any minerals at all. This means the water rushing over the white stones slips through the landscape without picking up heavy dissolved salts, sediment, or minerals. It is a glimpse of a fast-flowing, rocky hydrological system that naturally prevents heavy sediment accumulation.

Water Salinity:

Water salinity at Sada Pathor is surprisingly low and completely safe between 21.90 mg/L and 60.43 mg/L at all three sampling sites. That's way below the 500 mg/L mark set by the World Health Organization for excellent quality freshwater. So, despite the rocky, mineral-heavy terrain, these streams really are pure freshwater. There are no traces of industrial waste or heavy salt leaching from the riverbed rocks. It shows a highly dynamic, actively flushed water system where the continuous flow of the river keeps the water clean and free of salt buildup.

Water pH:

Water at Sada Pathor stands out for its slightly alkaline, healthy profile. When test the pH at three different spots, the numbers come in between 7.33 and 7.88. That's right on target the World Health Organization's recommended range for drinking water is 6.5 to 8.5, so Sada Pathor fits comfortably inside it. It don't find the acidity of forest streams here, where decaying leaves drag the pH down. This river gets washed by minerals from rocks, which keeps the water balanced. It's the kind of chemistry is wanted for aquatic life: neutral, clean, and stable. The ecosystem thrives under these conditions, with everything pointing to a river that's not just safe for people, but a healthy home for fish and plants too.

Electrical Conductivity (EC) of Water:

Water at Sada Pathor barely conducts electricity its EC hovers between 39.57 and 116.00 µS/cm across the spots where samples were taken. That's well below the WHO's "excellent" threshold of 400 µS/cm, which means this water has very few dissolved inorganic salts.

This low EC perfectly matches up with the really low levels of TDS and minimal salinity. Basically, this crystal-clear river water is almost completely free of heavy metals or chemical runoff. It's the kind of pure water found in a fast-flowing, rocky landscape where the water doesn't sit long enough to absorb heavy minerals from the ground. So, the water remains lightly mineralized about as clean and naturally flushed as an open river gets.

Overall Water Condition:

Sada Pathor's water is remarkably clean and perfectly balanced. The numbers tell the story of a pristine, fast-flowing riverine system. By World Health Organization standards, there are very few dissolved minerals or salts floating around. Total Dissolved Solids peak at just 56 mg/L, salinity sits below 61 mg/L, and electrical conductivity doesn't even cross 116 µS/cm. All of these are well within what's considered excellent, making it clear there's no heavy pollution or dangerous chemical runoff.

What's even better is the pH, which sits right in the optimal WHO range of 7.33 to 7.88. This perfectly neutral to slightly alkaline water is characteristic of a clean, mineral-washed rocky river. In the end, Sada Pathor's water is crystal clear, naturally flushed, and stands as a prime example of excellent open-water purity.

Table 15 Water Quality, Sada Pathor

Place	S\N	Lat	Long	Water Conductivity (µS/cm)	Water pH	Water Salinity (mg/L)	TDS (mg/L)
Sada Pathor	Point 1	25.15576	91.7494	116	7.33	60.43	56.43
	Point 2	25.16639	91.75109	109	7.88	59.23	54.77
	Point 3	25.17166	91.75101	39.57	7.71	21.9	19.53

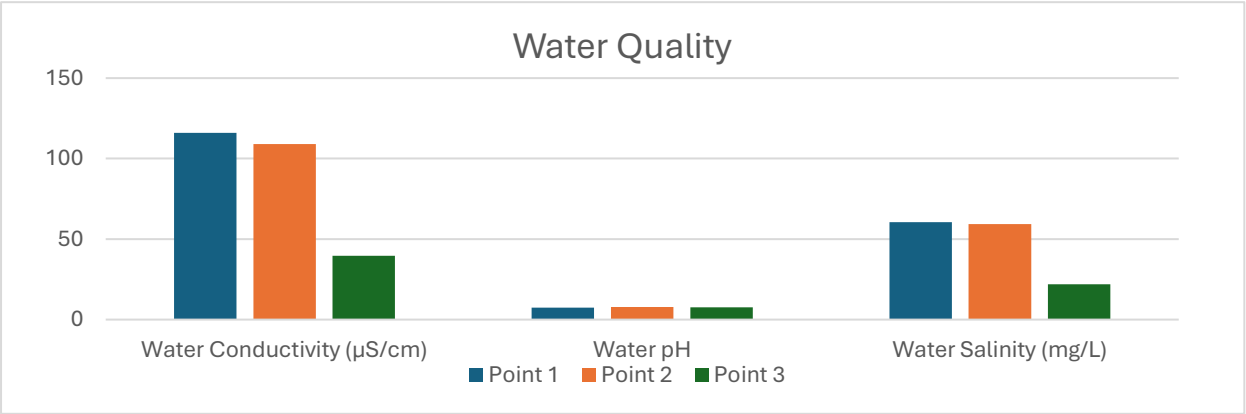


Figure 45 Water Quality, Sada Pathor

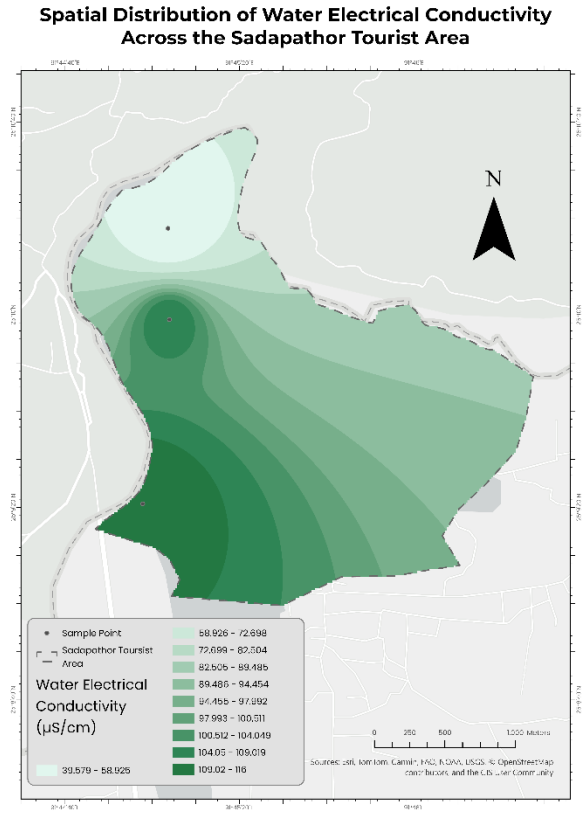


Figure 46 Water EC, Sada Pathor

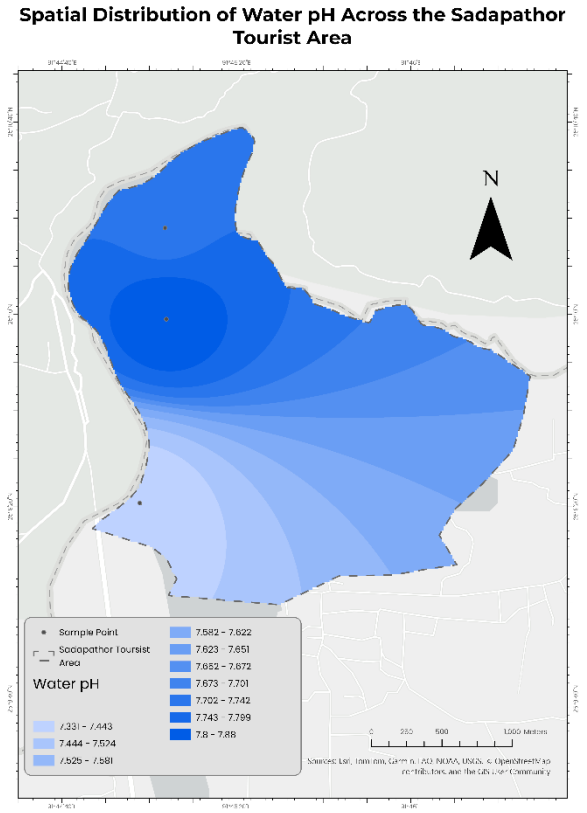


Figure 47 Water pH, Sada Pathor

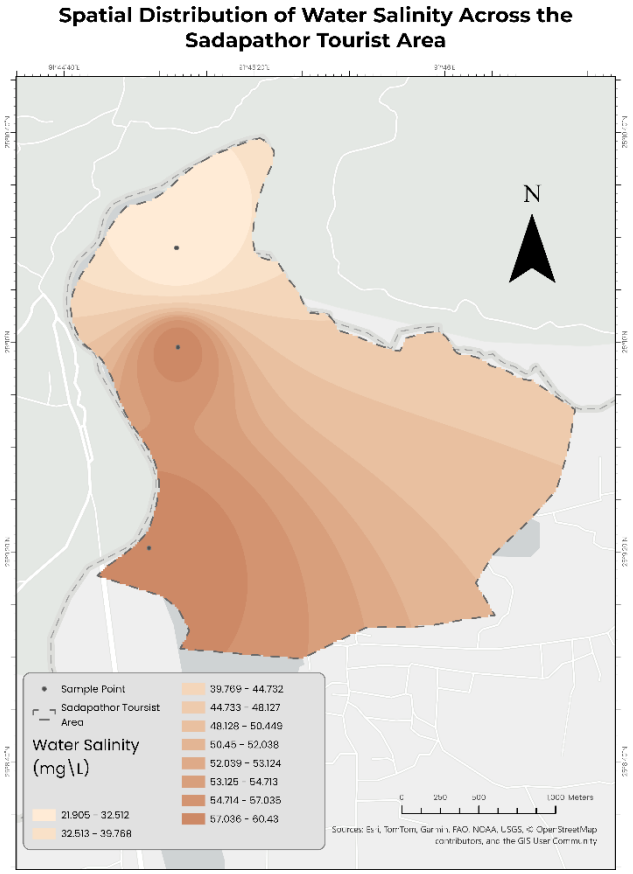


Figure 48 Water Salinity, Sada Pathor

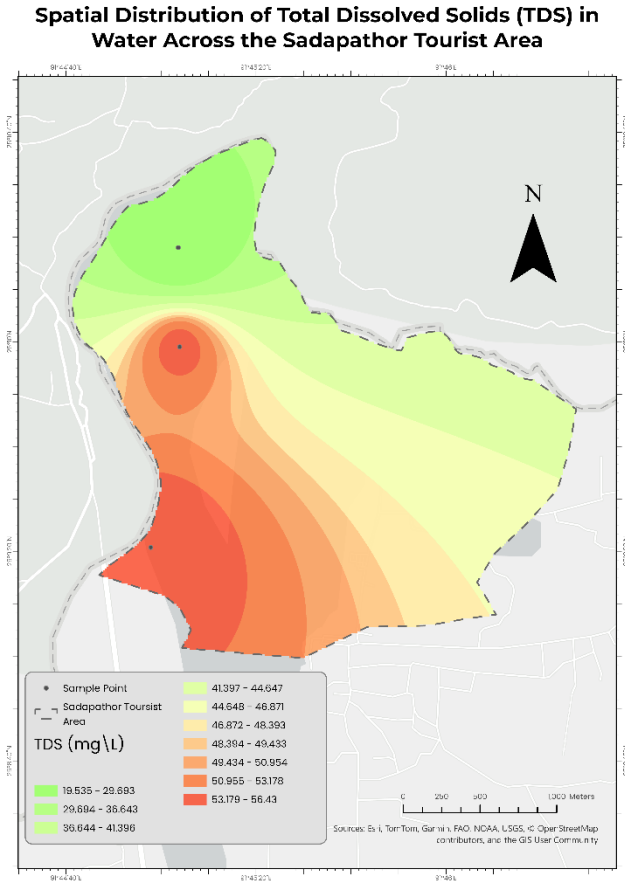


Figure 49 TDS, Sada Pathor

Soil Condition

Soil pH and Moisture:

Soil pH at Sada Pathor falls between 5.90 and 6.83, depending on where is the sample. That means most of these spots land in the ideal range most experts say anything from 6.0 to 7.5 works best for healthy soil. So, despite the tough environment on the surface, the soil hasn't been stripped bare or damaged by chemicals. It's actually in decent shape.

Moisture, on the other hand, tells a very different story. Readings jump from a bone-dry 10.67% all the way to 35%. Fertile soil usually holds somewhere between 20% and 50% water. At Sada Pathor, those super dry patches sitting stubbornly around 10% make it clear just how unforgiving the sun can be. There's no overhead shade, no leafy ground cover, nothing to slow down evaporation. The only places to be found any real moisture, say the surprisingly high 35%, are right next to the riverbed. Everywhere else, the soil fights and fails to keep water under the relentless 40°C heat.

Soil Electrical Conductivity (EC):

Sada Pathor's soil tells a story through its electrical conductivity it swings from a low 38.33 µS/cm all the way up to 199.33 µS/cm, depending on where the sample. That's nowhere near the 2,000 µS/cm mark that signals saline soil in agronomy and soil science. So, looking at completely non-saline soil here; salts just aren't an issue.

Why such a big jump between spots? That unevenness points to the patchwork nature of the land. The rocky, sandy ground doesn't get steady nutrient cycling from a thick cover of organic debris. Instead, rare river floods and scattered mineral pockets shape the chemistry, creating little pockets of higher conductivity while most of the soil stays clean and sandy.

Soil Texture:

Sada Pathor's soil is loose and sandy, just what would expect in a riverbed packed with rocks. When we ran the ribbon length test, the numbers dropped all the way to 0.0 mm at two spots meaning the soil couldn't stick together at all. That's pure sand or gravel, no clay minerals, almost no water retention. At another spot, we hit 28.83 mm, which hints at a bit of light loam sediment the river might've dropped off, holding just enough clay to keep a bit of shape. So, the place is mostly covered in loose sands and rocks that drain quickly, not the kind of soil call fertile or cohesive.

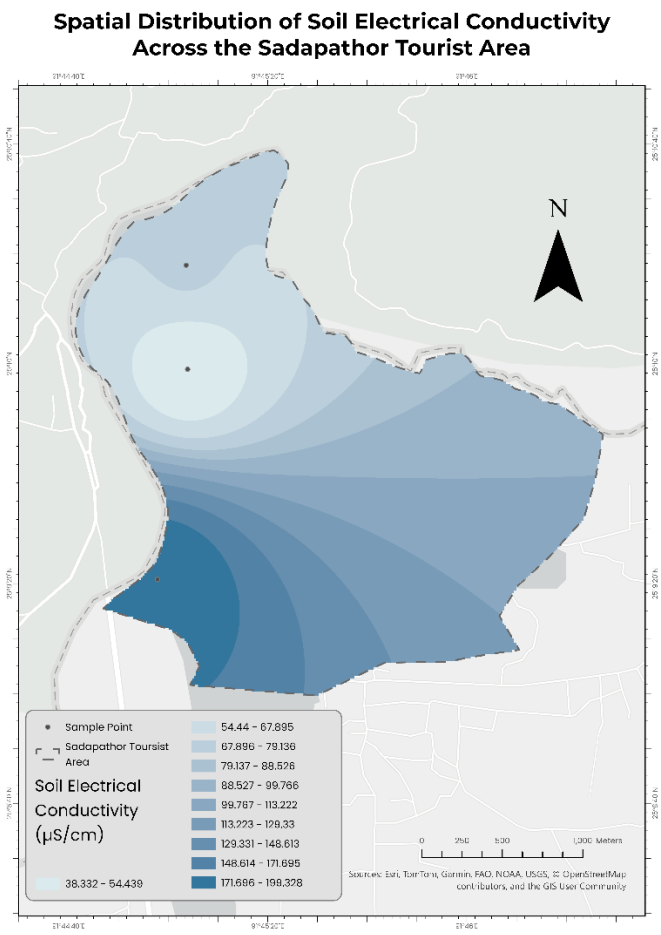


Figure 53 Soil EC, Sada Pathor

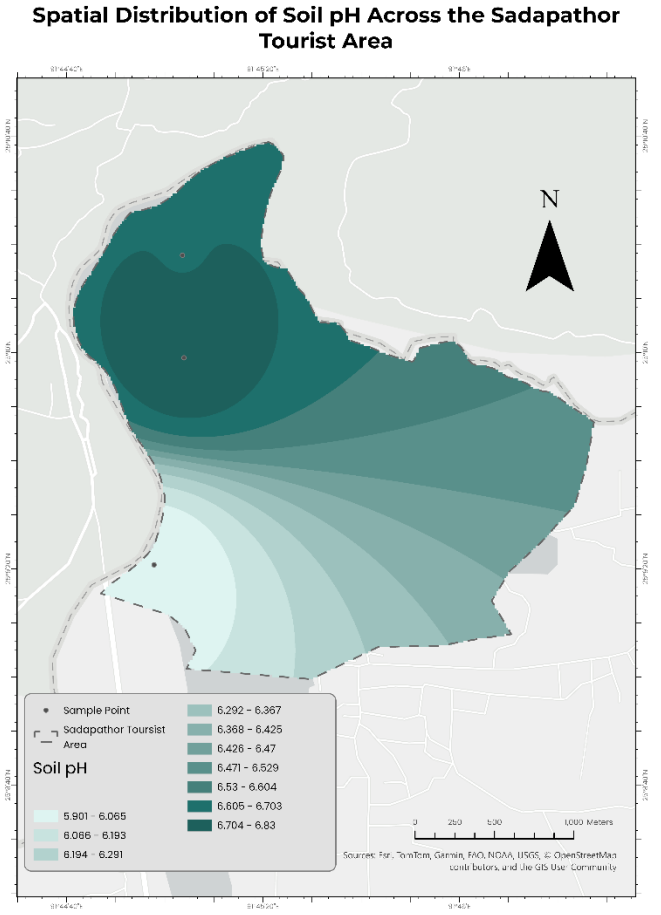


Figure 52 Soil pH, Sada Pathor

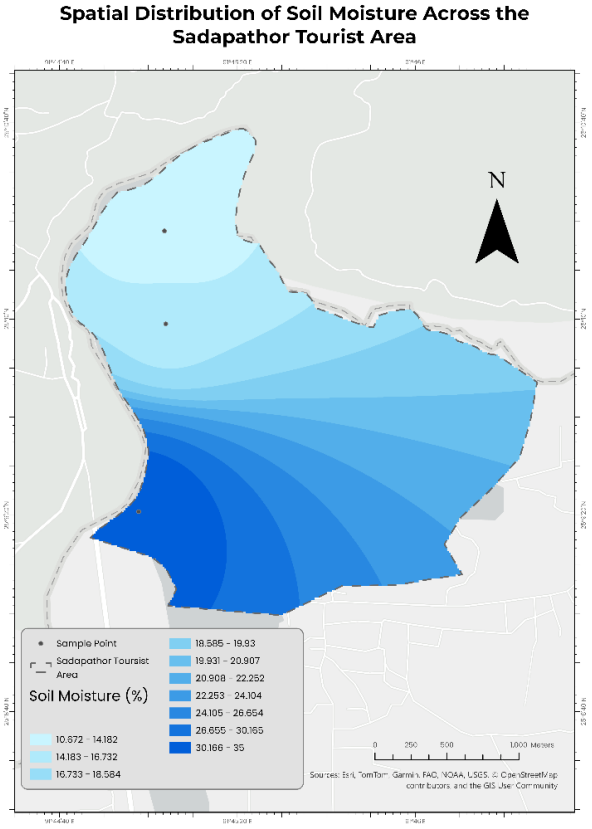


Figure 50 Soil Moisture, Sada Pathor

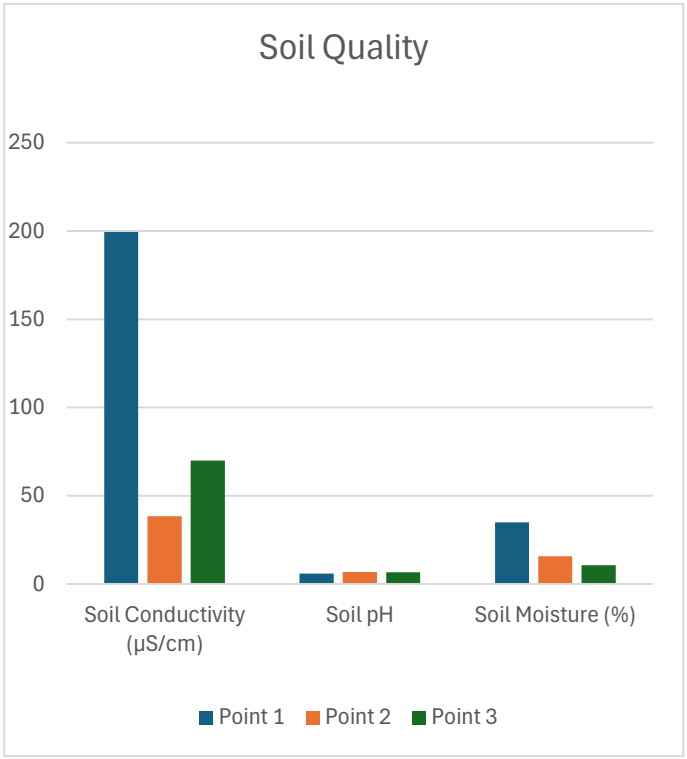


Figure 51 Soil Quality Graph, Sada Pathor

Table 16 Soil Quality, Sada Pathor

Place	S\N	Lat	Long	Soil Conductivity (µS/cm)	Soil pH	Soil Moisture (%)	Soil Ribon (mm)
Sada Pathor	Point 1	25.15576	91.7494	199.33	5.9	35	28.83
	Point 2	25.16639	91.75109	38.33	6.83	15.67	0
	Point 3	25.17166	91.75101	70	6.7	10.67	0

Overall Soil Condition:

Sada Pathor's soil says it all: harsh, rocky, shaped by the river. The pH? Just right, sitting between 5.90 and 6.83. Beyond that, things get rough. There's no mulch, no canopy, nothing to protect the landscape from relentless 40°C heat. The ground dries out fast since moisture barely sticks around, dropping to about 10.67% in the sandiest spots. So, good pH and it's almost free from any salt electrical conductivity sticks to the low end, between 38 and 199 µS/cm. But the soil texture tells a tougher story. Ribbon tests hit 0.0 mm, so it's almost pure sand and rock, missing the clay that would help hold onto nutrients and water. It's classic river terrain: clean, no salinity issues, but loose, exposed, and struggling for hydration under the sun.

Overall Environmental Condition:

Sada Pathor is an extreme, intense, but highly pure riverine ecosystem. It stands in total contrast to shaded forests; this environment is shaped by punishing heat and highly reflective rocky terrain. The microclimate is aggressively hot consistently hovering around 40.5°C and paired with moderate humidity (around 60%) and zero wind, the air feels completely stagnant and stifling. Because there is no forest canopy to act as a natural filter, the air carries a moderate amount of dust (PM 10 up to 22.67 µg/m³), giving it a harsher, drier feel.

From a hydrological perspective, Sada Pathor comes close to ideal. The river rushes past with a pH right inside the WHO's recommended range for drinking water between 7.33 and 7.88. It's clear as glass, with hardly any dissolved minerals; TDS always stays below 57 mg/L. Salinity is almost nonexistent. This is as pure and neutral as freshwater gets. The soil tells a similar story. It's not saline and its pH is just as balanced, but there's a catch: the soil is mostly sand, loose and quick to dry out. Under the open sun, it can lose moisture fast sometimes dropping to just 10%.

Ultimately, Sada Pathor presents a beautifully rugged but extreme environmental picture. It is a sun-baked, windless, and intensely hot landscape, yet perfectly balanced by some of the cleanest, fastest-flowing open water in the region, creating a stark but pristine natural environment.

4.3 Site Based Observation

Each place stands out for its own flora, wildlife, and geology, maybe a rare tree line here, a cluster of fossils there, or unexpected layers of rock. These details give a site its character and spark real scientific curiosity. That's what draws researchers in those distinctive features you can't find anywhere else.

4.3.1 Lawachara National Park

Flora

Taxonomic Classification of Observed Species

Sample 1: *Musa acuminata* (Wild Banana)

Kingdom: *Plantae*

Division: *Tracheophyta*

Class: *Liliopsida* (Monocots)

Order: *Zingiberales*

Family: *Musaceae*

Genus: *Musa*

Species: *Musa acuminata*



Figure 54 Musa acuminata

Sample 2: *Melastoma malabathricum* (Malabar Melastome)

Kingdom: *Plantae*

Division: *Tracheophyta*

Class: *Magnoliopsida* (Dicotyledons)

Order: *Myrtales*

Family: *Melastomataceae*

Genus: *Melastoma*

Species: *Melastoma malabathricum*



Figure 55 Melastoma malabathricum

Sample 3: *Trametes versicolor* (Turkey Tail Fungus)

Kingdom: *Fungi*

Phylum: *Basidiomycota*

Class: *Agaricomycetes*

Order: *Polyporales*

Family: *Polyporaceae*

Genus: *Trametes*

Species: *Trametes versicolor*



Figure 56 Trametes versicolor

Sample 4: *Gmelina arborea* (Gamhar)

Kingdom: *Plantae*

Division: *Tracheophyta*

Class: *Magnoliopsida*

Order: *Lamiales*

Family: *Lamiaceae* (formerly *Verbenaceae*)

Genus: *Gmelina*

Species: *Gmelina arborea*



Figure 57 Gmelina arborea

Sample 5: *Tectaria sp*. (Pahari Fern)

Kingdom: *Plantae*

Division: *Polypodiophyta* (Ferns)

Class: *Polypodiopsida*

Order: *Polypodiales*

Family: *Tectariaceae*

Genus: *Tectaria*

Species: *Tectaria sp*.



Figure 58 Tectaria sp.

Sample 6: *Tectona grandis* (Segun / Teak)

Kingdom: *Plantae*

Division: *Tracheophyta*

Class: *Magnoliopsida*

Order: *Lamiales*

Family: *Lamiaceae*

Genus: *Tectona*

Species: *Tectona grandis*



Figure 59 Tectona grandis

Sample 7: *Dipterocarpus turbinatus* (Garjan)

Kingdom: *Plantae*

Division: *Tracheophyta*

Class: *Magnoliopsida*

Order: *Malvales*

Family: *Dipterocarpaceae*

Genus: *Dipterocarpus*

Species: *Dipterocarpus turbinatus*



Figure 60 Dipterocarpus turbinatus

Sample 8: *Phrynium pubinerve* (Packing Leaf / Pitha Pata)

Kingdom: *Plantae*

Division: *Tracheophyta*

Class: *Liliopsida* (Monocots)

Order: *Zingiberales*

Family: *Marantaceae*

Genus: *Phrynium*

Species: *Phrynium pubinerve*



Figure 61 Phrynium pubinerve

Sample 9: *Amomum sp*. (Wild Cardamom)

Kingdom: *Plantae*

Division: *Tracheophyta*

Class: *Liliopsida* (Monocots)

Order: *Zingiberales*

Family: *Zingiberaceae*

Genus: *Amomum*

Species: *Amomum sp*.



Figure 62 Amomum sp.

Fauna

Lawachara National Park isn't just a patch of green on the map it's the heart of Bangladesh's wildlife. What really makes it special? This is the last home for the Western Hoolock Gibbon, which, by the way, happens to be the country's only ape. In the trails, capped langurs, Phayre's leaf monkeys, a few rhesus macaques, and pig-tailed macaques swinging through the branches. One time, I spotted a Malaysian Black Squirrel darting up a tree so quick you could almost miss it.

But it's not just about the primates. The park brims with other mammals barking deer slip through the undergrowth, wild boar root around in the soil, civets lurk in the shadows, and squirrels and bats seem to pop up everywhere. Birdsong never stops here. Some birds nest all year, others drop in just for a season, but all of them bring bursts of color and life up in the canopy, down in the bushes, and right at your feet. Head down to the forest floor or follow a stream, and lizards, snakes, frogs, and toads going about their business. And there's a whole hidden world of invertebrates the quiet workers pollinating flowers, breaking down leaves, and keeping the whole place humming along. Lawachara's wild heart beats because of all these creatures, big and small.



4.3.2 Jaflong Zero Point

Geomorphology and Geography

Location and Setting

Jaflong Zero Point sits right on the northern edge of Bangladesh, in Gowainghat Upazila, Sylhet District, touching the border with India's Meghalaya state. The Dawki River locals call it the Piyain flows alongside, marking a busy spot for people crossing between Bangladesh and India. Tourists flock here, drawn by the striking landscape, the lively river, and the stone collection that's become part of local life. Jaflong has earned its place as one of the most popular places to visit in northeastern Bangladesh.



Climate

The area sits between a humid subtropical and a tropical monsoon climate. Being so close to the Meghalaya Plateau one of the rainiest places on the planet it gets drenched during the monsoon season. All that rain shapes the way rivers flow, how sediment moves, and even how the land looks over time.

The Dauki Fault

Jaflong is especially important for studying the Dauki Fault, a major earthquake-prone fault line in Meghalaya, India. This fault follows the Dauki (Piyain) River and, at Jaflong, you can see it clearly. It lies right at the base of the Shillong hills, facing the edge of the Meghalaya Plateau. If you look at the accompanying picture, you'll see exactly where the fault cuts across, both from above and from the side, as seen from Zero Point.

Landform and Soils

The area sits between rivers and foothills, shaped by sediments flowing down from the Meghalaya hills. You see broad river channels, gravel bars, sandy floodplains, and gentle slopes at the foothills all classic signs of this landscape.

The soil here doesn't hold much water. It's sandy, gravelly, and alluvial, with barely any organic matter. Because it's geologically young, seasonal floods keep reshaping it, wash away layers, and pile new ones on top. When monsoon rains hit, the rivers swell. The banks erode, fresh sediments settle,

and the river itself can shift course. Once the dry season rolls in, riverbeds are laid bare, and people flock there to collect stones.

Geology

Jaflong sits right where the Meghalaya Plateau meets the Surma Basin, a spot shaped by ancient sedimentary rocks sandstone, limestone, and shale that came from the uplifted Meghalaya plateau during the Tertiary period.



The river does most of the heavy lifting here. It carries loads of boulders, pebbles, and gritty sand, all eroded off tough rocks upstream in India, and dumps them across the riverbed and floodplain. That's why Jaflong collects so much stone and gravel it's basically a natural storage zone. The upstream rocks are full of joints and fractures, and when heavy rains hit, these spots weather fast, breaking down into material that gets swept downstream.

Stone collection and processing dominate the local economy. People rely on it, but it's not without consequences. The activity is essential, yet it's doing real environmental damage. Jaflong's landscape grabs your attention clear blue water, piles of white and gray stones, and lush green hills from Meghalaya creating a dramatic contrast.

Tourists flock here, drawn by the scenery and the chance to see the international border up close. But it's a double-edged sword: unregulated tourism, mining the riverbed, and careless waste disposal are eroding habitats and putting too much strain on the river ecosystem.

Rocks and Minerals (Jaflong)

Sample 1: Sandstone Pebble (Cream-Yellow)

Property	Description
Rock Type	Sedimentary (clastic)
Color	Cream to yellow
Texture	Fine-grained, sandy
Structure	Layered (bedded) or massive
Main Minerals	Quartz, with silica or calcite cement
Diagnostic Feature	Cream-yellow color + fine, uniform, gritty grains



Figure 63 Rock Sample 1

Sample 2: Granodiorite

Property	Description
Rock Type	Igneous (intrusive)
Color	Pink and white with dark specks
Texture	Coarse-grained (phaneritic)
Structure	Massive
Main Minerals	Plagioclase, quartz, orthoclase, biotite, hornblende
Diagnostic Feature	Speckled pink-white look with more plagioclase than K-feldspar



Figure 64 Rock Sample 2

Sample 3: Diorite

Property	Description
Rock Type	Igneous (intrusive)
Color	Gray-green, salt-and-pepper
Texture	Coarse-grained (phaneritic)
Structure	Massive
Main Minerals	Plagioclase, hornblende, biotite
Diagnostic Feature	Salt-and-pepper look with little or no quartz



Figure 65 Rock Sample 3

Sample 4: Ferruginous Sandstone

Property	Description
Rock Type	Sedimentary (clastic)
Color	Maroon-purple to reddish-brown
Texture	Sandy, medium-grained
Structure	Layered (bedded) or massive
Main Minerals	Quartz, with iron oxide (hematite) cement
Diagnostic Feature	Deep maroon-purple color from iron-rich cement



Figure 66 Rock Sample 4

Sample 5: Pink Quartzite

Property	Description
Rock Type	Metamorphic
Color	Pale pink
Texture	Fine to medium-grained, interlocking (granoblastic)
Structure	Massive
Main Minerals	Quartz, with trace iron or manganese for the pink tint
Diagnostic Feature	Dense, hard rock with sparkly interlocking quartz grains



Figure 67 Rock Sample 5

4.3.3 Sada Pathor, Volaganj

Geomorphology and Geography

Location and Setting

Volaganj Sada Pathor sits in Companiganj Upazila, Sylhet District, not far from the India-Bangladesh border near the state of Meghalaya. The site stretches along the Dhalai River, where the riverbed and floodplain are packed with striking white stone deposits the “sada pathor” that gives the area its name. Locals and travelers know it as one of northeastern Bangladesh’s most remarkable geological features and a major tourist destination.



Climate

This region sits under a humid, tropical monsoon climate think heavy, drenching rains, especially when the monsoon rolls in. Upstream, the Meghalaya hills catch the brunt of these downpours, sending water rushing down and shaping everything in its path. River discharge spikes, stones tumble in, and the whole Volaganj landscape shifts.

Landform and Soils

Out here, riverine gravel plains and thick stone beds stretch across the land. Over time, the rivers have dumped loads of coarse sediment from the Meghalaya Plateau, leaving behind sweeping channels, wide patches of exposed boulders, long gravel bars, and gentle foothills.

Soil, There’s hardly any. The rivers constantly flood and stir things up, so there’s barely a chance for proper soil to form. Where dirt does manage to hang on, it’s rough and gritty more sand and rocks than anything, with barely any organic matter. During the monsoon, the river turns wild, flooding the stone beds and carving new paths through the land. When the rains finally ease and the dry season takes over, the water pulls back, revealing vast stretches of gleaming white stones that have become the area’s signature look.

Geology

Volaganj Sada Pathor sits right where the Meghalaya Plateau fades into the Sylhet Basin. The landscape up in the Meghalaya hills looks rugged, built from tough rocks like sandstone, limestone, and quartz. Over time, heavy monsoon rains and the steady rise of the plateau break up these rocks. You end up with boulders, pebbles, and coarse sand pouring down the hills.

The Dhalai River does the heavy lifting here, picking up that debris and dropping it at Volaganj. What you get is a wide, stony plain white rocks everywhere. Those clean fractures upstream help split the blocks apart, adding to the sea of pale stones you see when you arrive.

All those stone beds give Volaganj its signature look, especially when the water's low. The place comes alive visually: gleaming white stones, crystal-clear river, and lush green hills all around. It draws in tourists looking for something dramatic and beautiful.

But Volaganj isn't just a sightseeing spot. Lots of people depend on collecting the riverbed stones for construction it's part of their livelihood. That said, the surge of visitors and loose regulation on extraction have started to cause trouble. The riverbed gets chewed up, habitats get pushed aside, and erosion creeps in faster than before. It's a classic story of natural beauty facing off against human pressure.

Rocks and Minerals (Sada Pathor)

Sample 1: Magnetite

Property	Description
Mineral Type	Iron oxide
Color	Black to grayish-black
Luster	Metallic to submetallic
Streak	Black
Hardness	5.5 to 6.5
Diagnostic Feature	Strongly magnetic



Figure 68 Rock Sample 6

Sample 2: Granite

Property	Description
Rock Type	Igneous (intrusive)
Color	Pink, white, or light gray with dark specks
Texture	Coarse-grained (phaneritic)
Structure	Massive
Main Minerals	Quartz, feldspar, mica
Diagnostic Feature	Visible interlocking crystals of mixed colors



Figure 69 Rock Sample 7

Sample 3: Pink Feldspar

Property	Description
Mineral Type	Silicate (potassium feldspar)
Color	Pink to salmon
Luster	Vitreous to pearly
Hardness	6
Cleavage	Two directions at about 90°
Diagnostic Feature	Pink color + two cleavage planes at right angles

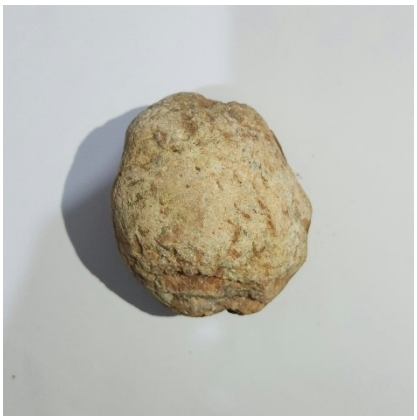


Figure 70 Rock Sample 8

Sample 4: Sandstone

Property	Description
Rock Type	Sedimentary (clastic)
Color	Tan, brown, red, or yellow
Texture	Sandy, medium-grained
Structure	Layered (bedded) or massive
Main Minerals	Quartz, with feldspar and cementing material
Diagnostic Feature	Gritty feel from cemented sand grains



Figure 71 Rock Sample 9

Sample 5: Milky Quartz

Property	Description
Mineral Type	Silicate (silicon dioxide)
Color	White to cloudy white
Luster	Vitreous (glassy), sometimes greasy
Hardness	7
Fracture	Conchoidal
Diagnostic Feature	Milky, translucent appearance + high hardness

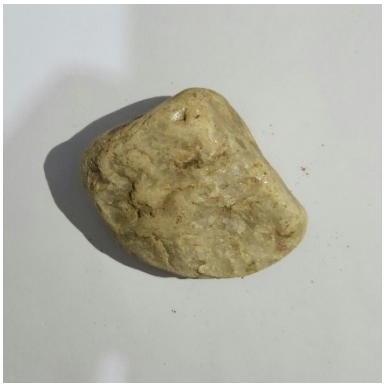


Figure 72 Rock Sample 10

4.3.4 Madhabkunda Eco Park

Geomorphology and Geography

Geological Characteristics and Rock Formation

At Madhabkundo Eco Park, a classic Tertiary sedimentary sequence right out in the open. Most of it comes from the Surma Group usually connected with the Bhuban and Boka Bil formations and reflects the geology of the Patharia Hill anticline. The cliff face lays it all out: sharp, horizontal or nearly flat layers that don't hide their structure. The rock bands switch back and forth. Those heavier, tough layers sticking out? That's sandstone. The more tucked in bands, a bit darker, are shale, siltstone, or mudstone, and they don't hold up quite as well against the elements.



Millions of years ago, deltaic and shallow marine environments laid down these sediments in the Bengal Basin. As time passed, the Indian and Eurasian plates collided, squeezing these sediment layers from the sides. That pressure pushed them up and folded them into the hills we see today. Now, the exposed cliff acts like a window into the past, showing layers that record the region's repeating patterns of deposition.

Geomorphological Characteristics

The landscape at the Madhabkundo waterfall owes its striking features to the region's geology. Structural control and the way different rocks break down shape the scenery. Sandstone stands firm against erosion, while the underlying shale crumbles much faster. Because of that, the cliff's layers wear away at different rates.

The waterfall stays dramatic because of differential erosion. The tough sandstone sits up top as a protective cap, but underneath, softer shale faces constant attack from the water and the humid air. While the shale crumbles away, the sandstone above loses support. Sooner or later, big slabs of it

break off and tumble down, often along deep cracks that run straight up and down. This keeps happening the shale gets eaten away, the sandstone falls, and the waterfall stays steep. That's why it doesn't mellow out into a slow, gentle slope; it holds onto that sharp, sheer drop, standing out as a classic geomorphological knickpoint.

Soil Formation (Pedogenesis)

Soil on this steep rock face doesn't have it easy. It's thin, always changing, and trace almost everything about it back to the bedrock and the tiny differences in climate right there on the cliff. Soil forms in the cracks and crevices places where the rock has weathered, or in layers of shale just deep enough to trap a little moisture and decaying debris.

Tropical downpours and constant waterfall spray beat down day in, day out. These speed up chemical weathering, breaking down the rock faster than drier spots. But deep, mature soil? It doesn't stand a chance. With the slope nearly vertical, loose material just washes away, carried by gravity and running water. So, end up with a thin, unstable layer lithosol, or entisol that barely clings to the rock.

The only thing keeping this young soil from disappearing is the tough network of roots from ferns, mosses, and broadleaf plants. These plants jam their roots right into the cracks, hanging on fiercely. When their leaves or stems die, they drop onto the rock, break down in no time, and release humic acids. All of this fuels more weathering and sets up a tight, nutrient rich cycle, creating a little ecosystem clinging to the edge of the cliff.

Site Uniqueness and Significance

This spot stands out for field study. First, the exposed stratigraphy here is a goldmine for anyone interested in geology. Researchers can walk right up and trace the layers of sandstone and shale in the Surma Basin. No need for drilling or piecing things together from small samples; just step back, and the rock tells its own history.

Then there's the geomorphology. Everything's happening right in front of eyes. The cliff is a living example of differential weathering and waterfall retreat, carving out new landforms. how structure shapes the landscape, etched straight into the stone.

The ecological angle is just as impressive. This vertical face supports a micro ecosystem won't be found on flat ground. Pioneer plants take root where they can, then secondary vegetation follows all on rock that's crumbling away, thanks to the humidity from the waterfall. It's nature adapting and evolving in real time, and the whole process relies on that unique microclimate.

Rocks and Minerals (Madhabkunda Eco Park)

Sample 1: Greywacke

Property	Description
Rock Type	Sedimentary (clastic)
Color	Dark grey, salt-and-pepper speckled
Texture	Fine to medium sand grains in a tan-grey matrix
Structure	Massive or bedded
Main Minerals	Quartz, feldspar, rock fragments, clay matrix
Diagnostic Feature	Dark speckled look + rounded, gritty, water-worn surface



Figure 73 Rock Sample 11

Sample 2: Ferruginous Siltstone

Property	Description
Rock Type	Sedimentary (clastic)
Color	Yellow-orange to rusty brown, with dark manganese-like patches
Texture	Very fine-grained (silt-sized), soft and friable
Structure	Thinly bedded; splits along bedding into flat, flaky pieces
Main Minerals	Quartz, clay, iron oxide
Diagnostic Feature	Leaves red iron-oxide powder on fingers + splits along bedding



Figure 74 Rock Sample 12

Sample 3: Mudstone

Property	Description
Rock Type	Sedimentary (clastic)
Color	Olive-tan
Texture	Smooth, uniform, no visible grains
Structure	Massive, with thin bedding cracks
Main Minerals	Clay minerals, fine silt



Figure 75 Rock Sample 13

Diagnostic Feature	Very soft (scratches with a finger) + smooth, grainless surface

Sample 4: Ferruginous Quartzite

Property	Description
Rock Type	Metamorphic (could also be chert)
Color	Rusty orange-brown iron coating, with a honey-yellow patch
Texture	Smooth, dense, slightly translucent and waxy in patches
Structure	Massive; rounded, polished pebble
Main Minerals	Quartz, iron oxide coating
Diagnostic Feature	Very hard and tough + rusty iron coating on a polished pebble



Figure 76 Rock Sample 14

Sample 5: Iron-Cemented Sandstone

Property	Description
Rock Type	Sedimentary (clastic)
Color	Brown with reddish-orange specks and grey-green grains
Texture	Fine, even, sandpaper-like
Structure	Massive; rounded, flattened stream pebble
Main Minerals	Quartz, with iron oxide cement
Diagnostic Feature	Hard, well-cemented grains with reddish-orange iron staining



Figure 77 Rock Sample 15

Chapter 05: Conclusion and Recommendations

5.1 Conclusion

Fieldwork in Sylhet gave the team an unfiltered glimpse into the messy, complicated terrain of northeastern Bangladesh. They didn't stay in one spot hills, riverbanks, wetlands, forests, even areas molded by people's hands, they explored it all. The theory from textbooks? It moved straight into real life with every muddy step. It's impossible not to notice how nature's rhythms and human decisions get tangled up, creating Sylhet's distinctive vibe. Shifts in terrain, sudden weather changes, soil underfoot, water clarity, the air as evidence. Nature and people, locked in, continuously shape the landscape and each other.

The observations really cleared things up. Vegetation, landforms, and hydrology they all shape the microclimate. Temperature swings, humidity, wind, and even the air's stability. Take Lawachara National Park, for example. The forests there keep things cooler, add moisture, calm the winds, and the soils look healthier. Natural vegetation does most of the work keeping environmental balance. But step into open or disturbed areas like Jaflong or Volaganj, and everything changes. Heat stress goes up, soils dry out, dust lingers, and the water is packed with dissolved solids and higher conductivity. Tourism, stone extraction, and land disturbances leave visible and measurable marks on the landscape.

But step into places like Jaflong or Volaganj open, disturbed, bustling with activity and that shift immediately. Heat stress ramps up, the soil dries out, dust lingers, and the water gets loaded with more conductivity and dissolved solids. Tourism, stone extraction, and land disturbances leave clear, measurable traces. The environment doesn't just react; it records every intrusion.

When checking water quality, things look decent overall. Most rivers and ponds have enough dissolved oxygen for fish and aquatic life to get by. But in extraction zones or busy tourist areas, salinity and dissolved solids start to rise, threatening river systems. Air and sound readings back this up even "protected" and "eco-tourism" spots can't escape dust and noise creep, making crystal clear the need for vigorous environmental management.

Soil tests told their own story. Low-lying and forested places had fine, moist soils that nurtured plants and kept ecosystems thriving. On the flip side, riverbeds and extraction sites showed rougher, more porous soils with lousy water retention prone to degradation. This shaped where vegetation could grow and whether land could bounce back after disturbance.

What really struck me about Sylhet is how wildly varied and delicate the environment is. Human activity and nature are tangled together everywhere. If the forests, rivers, and wetlands here are going to last, people need to plan land use with serious care, keep tourism in check, and put real effort into conservation not just talk about it. On the academic side, getting out in the field made everything

click. I got my hands dirty collecting data, learned to handle environmental instruments, practiced observing landscapes up close, and broke down the patterns I found. This wasn't just about memorizing concepts; it made me see plainly why sustainable management is crucial, especially somewhere as rich and sensitive as Sylhet.

5.2 Limitations of the Study

Although rigorous methodologies were applied during this study, several unavoidable constraints impacted the fieldwork and subsequent analysis. These limitations included:

1. Resource Constraints: Limited availability of essential analytical equipment restricted the broader scope of the study.

2. Time Limitations: Constrained field schedules hindered the ability to conduct extended observations and repeated spatial measurements.

3. Environmental Extremes: Extreme temperature conditions posed physical challenges and complicated in-situ data collection.

4. Weather Disruptions: Unpredictable weather, including sudden rainfall, intermittently interrupted scheduled field activities.

5. Transportation Hurdles: Logistical difficulties with transportation caused intermittent delays in the data collection process.

6. Coordination Challenges: Complex field logistics and scheduling constraints occasionally impacted overall operational efficiency.

7. Operational Delays: Limited prior field-handling experience with specific technical instruments resulted in minor operational slowdowns.

8. Calibration Constraints: The lack of immediate, on-site calibration facilities introduced potential constraints on absolute instrument precision.

9. Site Accessibility: Safely navigating to and surveying remote, geographically isolated sampling locations proved highly challenging.

5.3 Recommendations

To mitigate operational constraints and improve the efficacy of future fieldwork, the following recommendations are proposed:

1. Pre-Deployment Calibration: All analytical instruments should be rigorously tested, calibrated, and verified prior to field deployment to ensure baseline accuracy.

2. Equipment Redundancy: Potential mechanical and sensor faults should be minimized through routine maintenance and the availability of backup instruments.

3. Resource Preparedness: Sufficient auxiliary materials, spare batteries, and redundant devices must be carried to the field to prevent unexpected operational halts.

4. Workflow Optimization: Strict scheduling and a clear division of labor should be maintained among team members to maximize on-site efficiency.

5. Weather-Contingent Planning: Fieldwork schedules should be strategically aligned with favorable meteorological conditions to prevent disruptions from severe heat or sudden rainfall.

6. Measurement Stabilization: To ensure data accuracy, all sensor readings should only be recorded after allowing the device to stabilize for a minimum of one minute in the ambient environment.

7. Data Verification: All manually recorded values and field calculations should be immediately cross-checked to minimize the risk of human error.

8. Team Coordination: Clear and consistent communication must be maintained among all field personnel to avoid procedural overlaps or methodological confusion.

9. Safety Protocols: Appropriate protective gear and comprehensive first-aid equipment must be mandated and readily available during all on-site activities.

10. Future Research Scope: Subsequent studies should incorporate extended spatial sampling, seasonal comparative analyses, and advanced analytical tools to achieve a more comprehensive understanding of the local environmental conditions.

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Chapter 07: Appendices

7.1 Data Table

These are the finalized data after combining field results,

Table 17 Data Table

Parameter	Unit	Jaflong			Lawachara National Park			Sada Pathor		
		Sample 1	Sample 2	Sample 3	Sample 1	Sample 2	Sample 3	Sample 1	Sample 2	Sample 3
Lat	Decimal	25.18	25.18	25.19	24.31	24.32	24.33	25.16	25.17	25.17
Long	Decimal	92.01	92.02	92.01	91.79	91.78	91.78	91.75	91.75	91.75
Soil Conductivity	µS/cm	169	18.67	46.83	150.33	35.33	202.67	199.33	38.33	70
Soil pH	-	5.83	5.4	6.67	6.67	6.4	5.3	5.9	6.83	6.7
Soil Moisture	%	36.37	43.33	23.33	46.67	28.33	45	35	15.67	10.67
Soil Ribon	mm	4	23.67	0	0	17.33	69.33	28.83	0	0
Water Conductivity	µS/cm	115.47	37.5	75.13	36.03	33.6	42.66	116	109	39.57
Water pH	-	7.45	7.4	7.36	5.58	5.7	6.03	7.33	7.88	7.71
Water Salinity	mg/L	75.33	27.3	50.93	24.9	26.6	24.4	60.43	59.23	21.9
TDS	mg/L	115.2	39.96	75.4	20.7	20.67	19.34	56.43	54.77	19.53
Temp.	C	34.63	35.27	34.13	29	28.4	30.27	40.46	40.2	40.96
RH	%	50.6	43.33	44.3	78.67	81.9	74.8	61.13	63	58.45
Pressure	atm	1008.2	1006.26	1005.53	1003	1004.6	1005.77	1003.4	1003.6	1003.1
WS	m/s	1.07	0.9	0.5	0	0	0	0	0	0
WD	m\s	75.3	173.33	96.67	-	-	-	-	-	-
PM 2.5	-	4.33	3	3.33	7.3	7	5.3	12.33	18	14
PM 10	-	5.33	4	5.67	7.67	7.67	5.7	14.33	22.67	17.33

7.2 Pictures from Field Work





